Type 2 hypervisor
Wmware workstation



Type 2 hypervisor
Creating Virtual Machine

- **Topics to be Covered:**

  What is Type 2 Hypervisor

  Type of Type 2 Hypervisor

Type 2 Hypervisor

- It run at top of OS
- First you install OS and top of OS you install Wmware workstation / Wmw are fusion as application
- Using that application you create multiple VM

Type 2 Hypervisor



- **VMware Fusion**
- **Workstation**
- **VirtualBox**

VMware Fusion:

- This is the company's MacOS-focused offering, which lets Mac users run a large range of guest operating systems.
- Run window application in Mac OS

Workstation:

- VMware's Linux- and Windows-focused platform comes in two flavors
  - **Pro:**which is a paid version, The Pro version allows users to run multiple operating systems on a single PC
  - **Player**, which is free for personal use, Player only supports a single guest OS

- **Pro:** Paid, supports multiple VMs, offers advanced features.
- **Player:** Free, supports only one VM, ideal for personal or basic use.

VirtualBox



- A Type 2 hypervisor running on Linux, Mac OS, and Windows operating systems.

Type 2 hypervisor
Wmware workstation

Lab:
Creating Virtual Machine on Vmware Workstation

- **Topics to be Covered:**

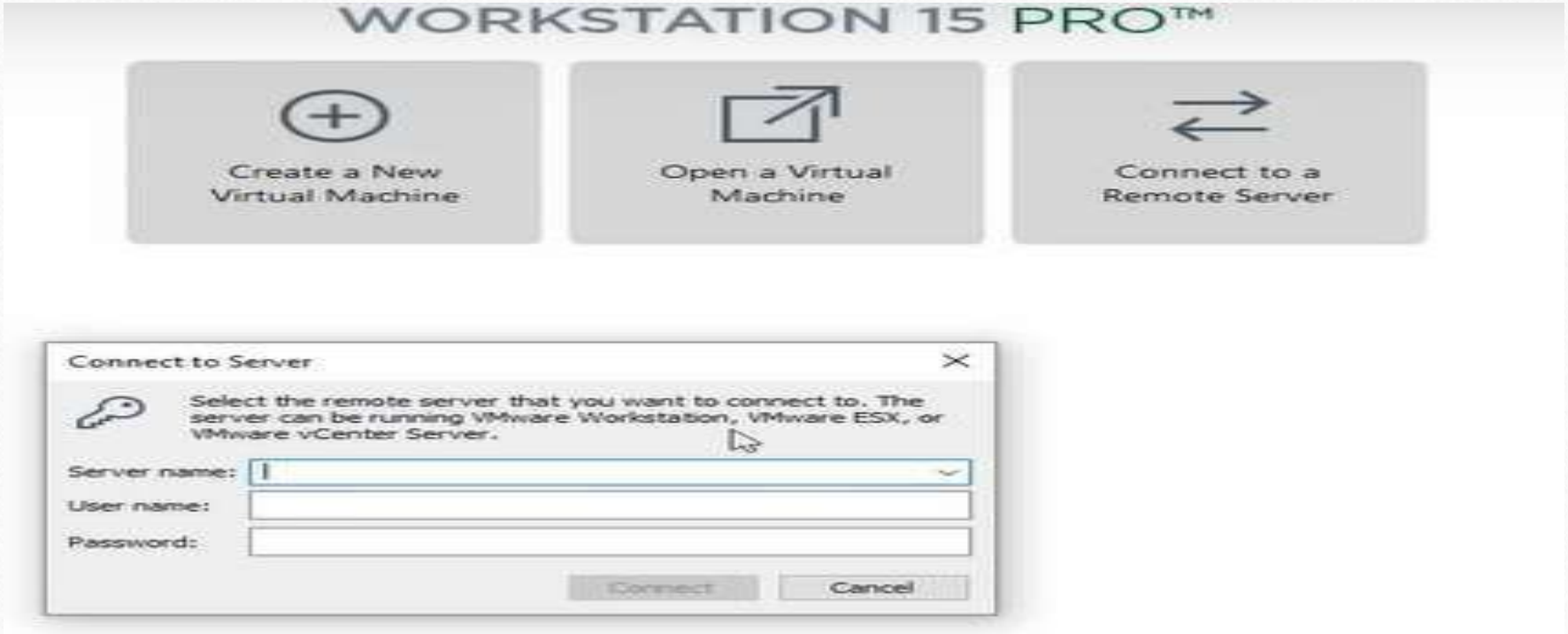
  Interface

  Create VM

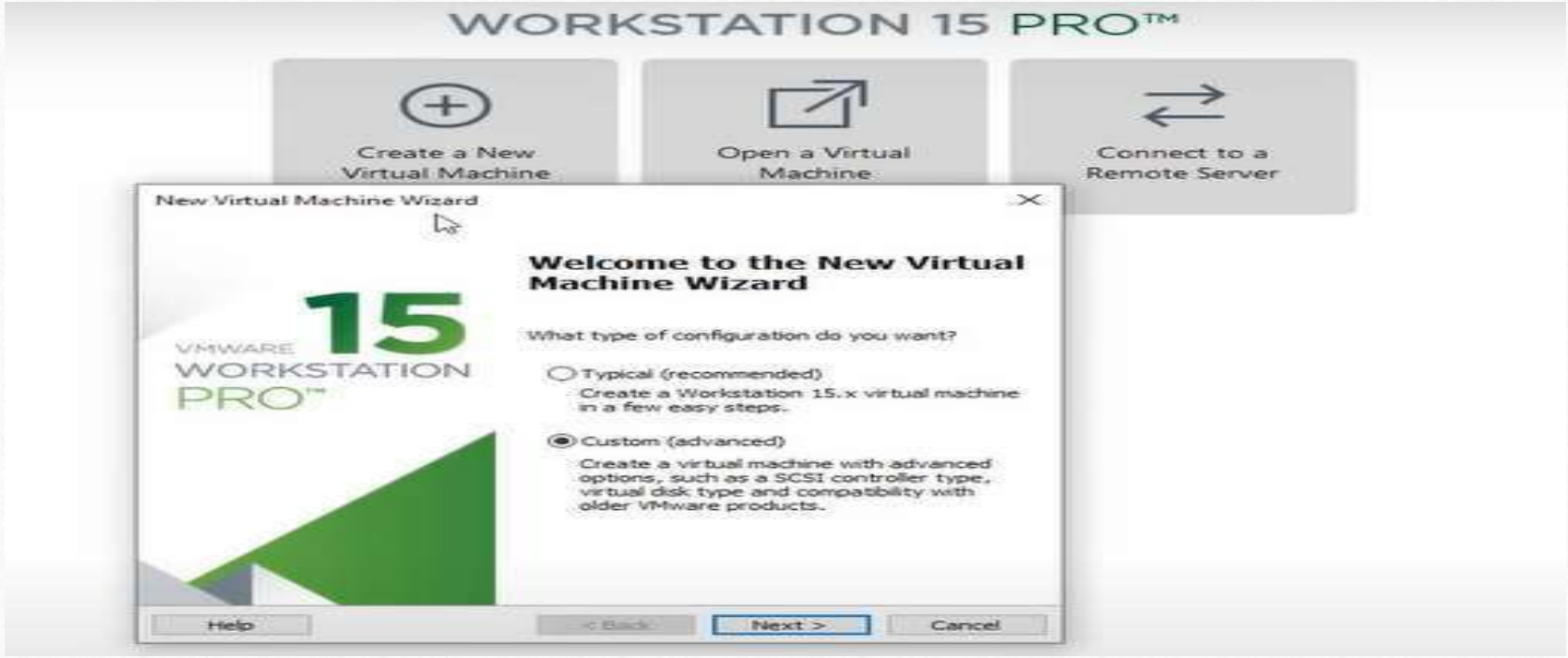
Interface


Type here to search
My Computer
KALI LINUX SHESH
CentOS 7 64-bit
Metasploitable2-Linux
win 7
Windows XP
Windows 10 (Shesh)
Win_SERVER 2019
RHEL
Shared VM
RHEL 7
C:\Users\INFOTECHSHESH\Documents\Virtual Machines\RHEL 7\RHEL 7.vmx
WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server

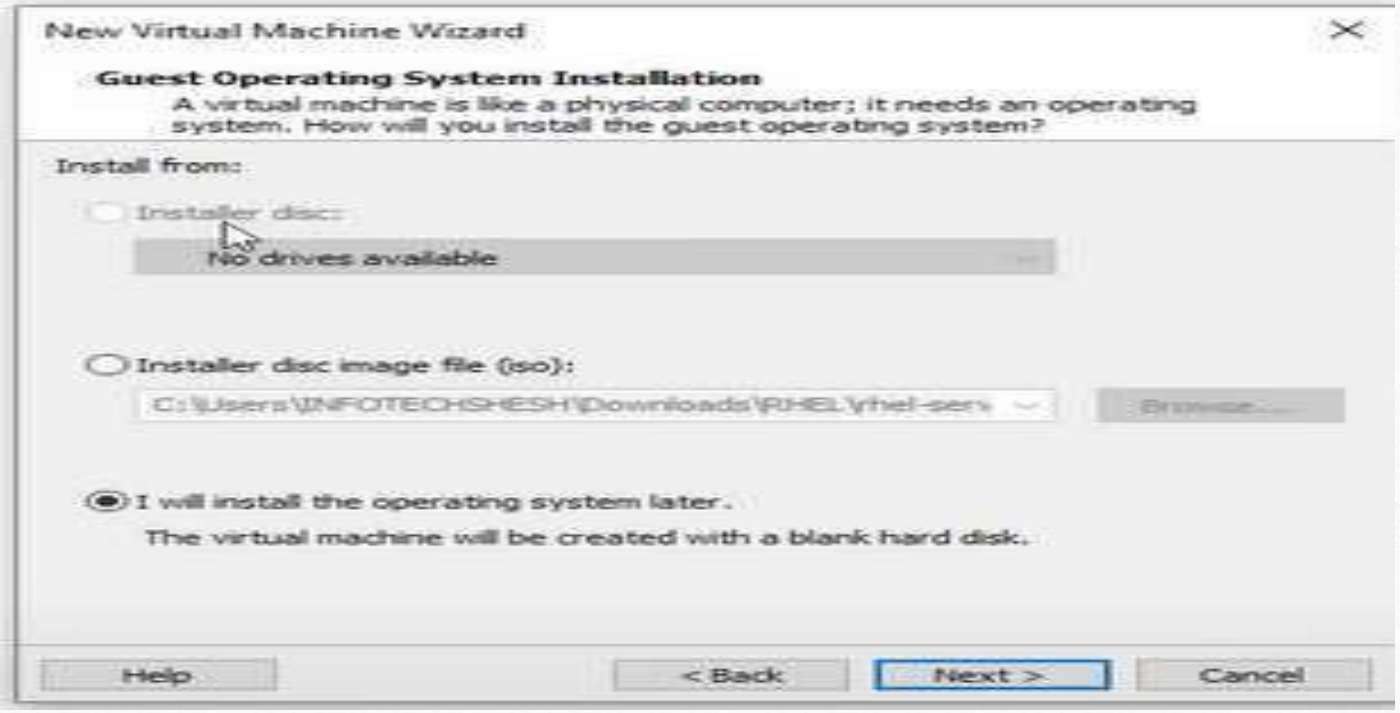
Interface


WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
Connect to Server
Select the remote server that you want to connect to. The server can be running VMware Workstation, VMware ESX, or VMware vCenter Server.
Server name:
User name:
Password:
Connect
Cancel

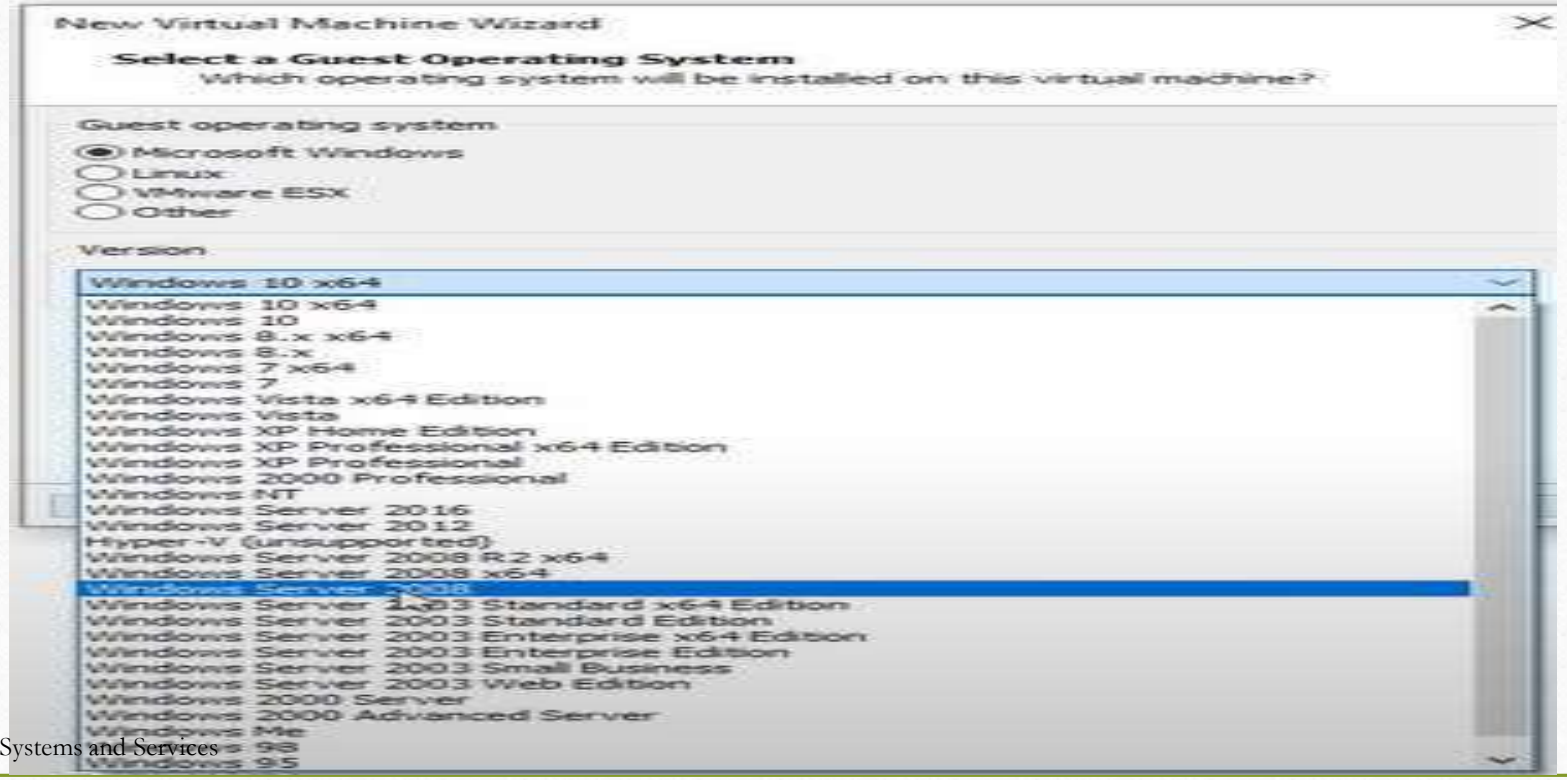
Create VM


WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
New Virtual Machine Wizard
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation 15.x virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
VMWARE 15 WORKSTATION PRO™
Help
< Back
Next >
Cancel

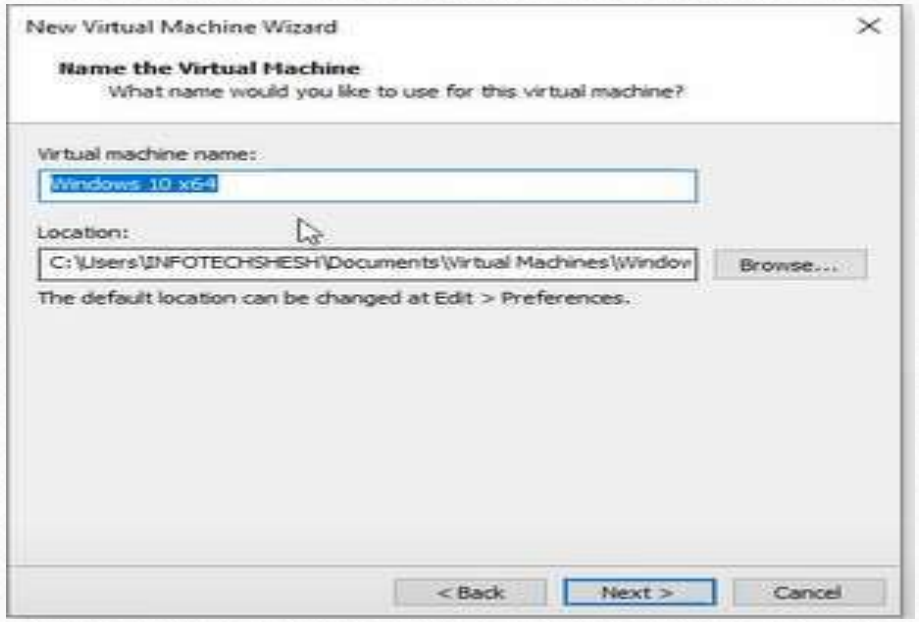
Create VM


New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

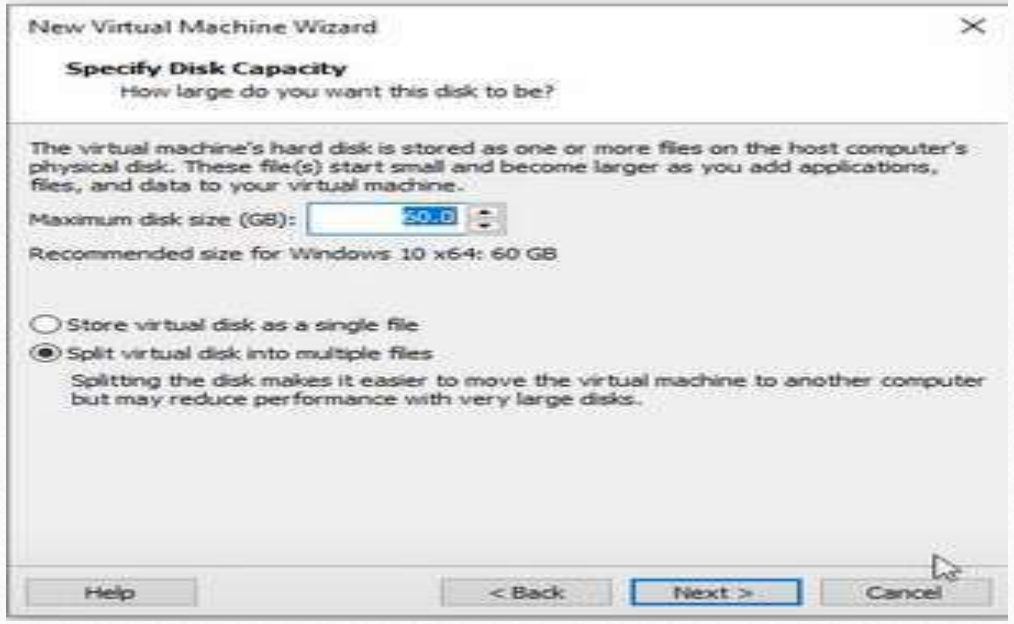
Create VM


New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Windows 10 x64
Windows 10 x64
Windows 10
Windows 8.x x64
Windows 8.x
Windows 7 x64
Windows 7
Windows Vista x64 Edition
Windows Vista
Windows XP Home Edition
Windows XP Professional x64 Edition
Windows XP Professional
Windows 2000 Professional
Windows NT
Windows Server 2016
Windows Server 2012
Hyper-V (unsupported)
Windows Server 2008 R2 x64
Windows Server 2008 x64
Windows Server 2008
Windows Server 2003 Standard x64 Edition
Windows Server 2003 Standard Edition
Windows Server 2003 Enterprise x64 Edition
Windows Server 2003 Enterprise Edition
Windows Server 2003 Small Business
Windows Server 2003 Web Edition
Windows 2000 Server
Windows 2000 Advanced Server
Windows Me
Windows 98
Windows 95

Create VM


New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Windows 10 x64
Location:
C:\Users\INFOTECHSHESH\Documents\Virtual Machines\Windov
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel

Create VM


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
50.0
Recommended size for Windows 10 x64: 60 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel

ESXi (Hypervisor)

ESXi (Hypervisor)

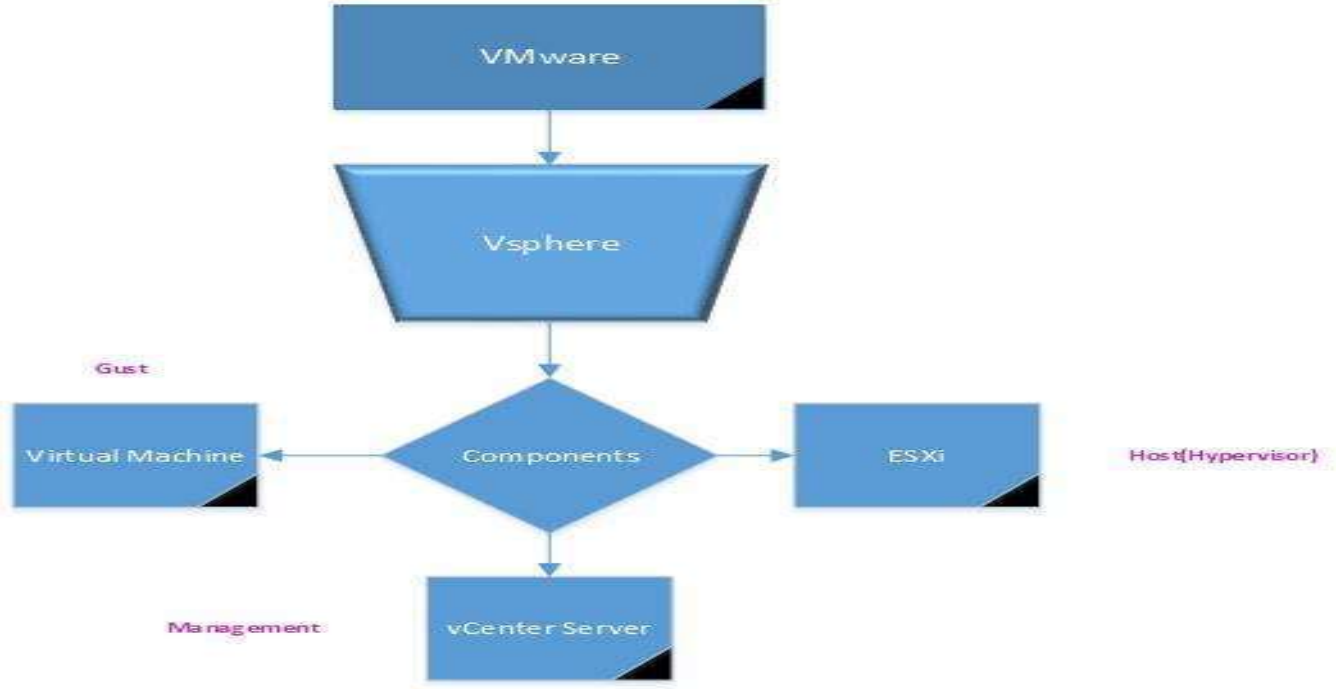
- **Topics to be Covered:**

  What is ESXI

What is ESXI

- It is Type 1 Hypervisor
- Version 6.7 vSphere (Current)

Component of Vsphere

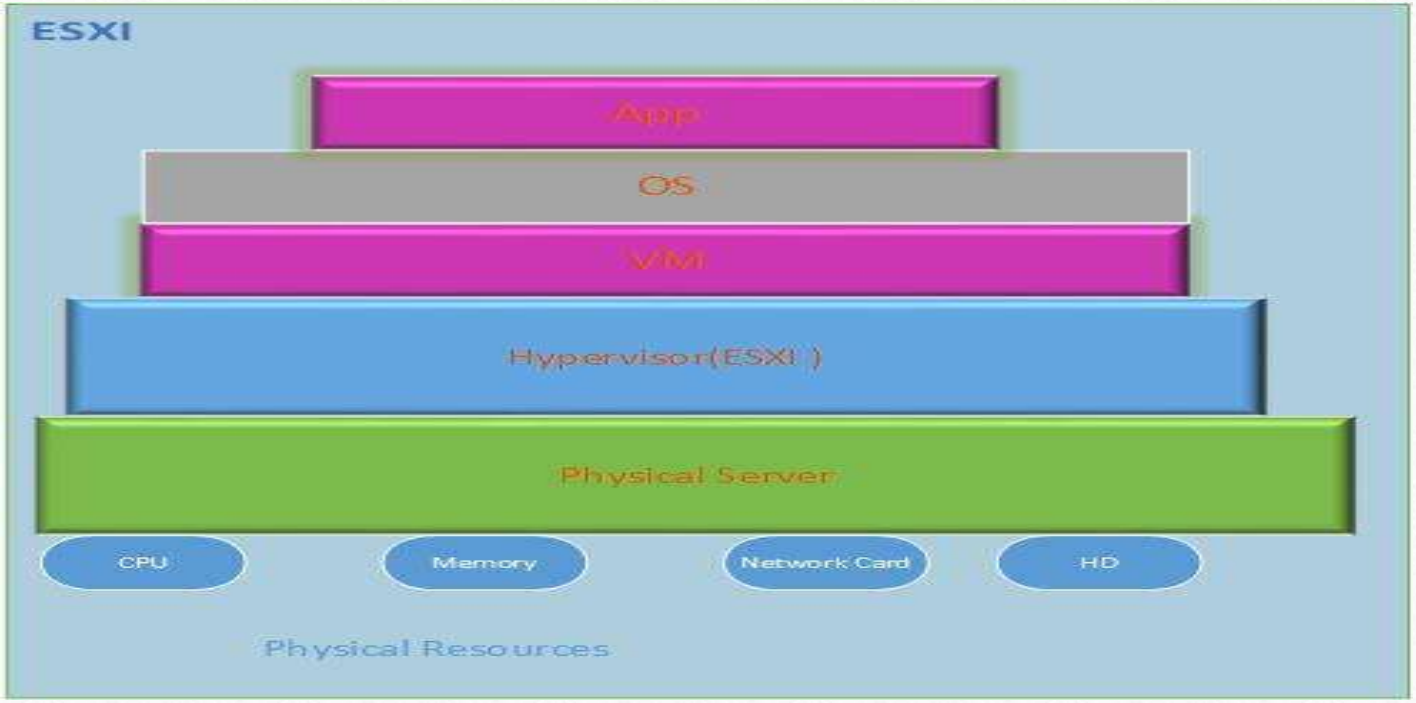


ESXI

ESXi ek house manager ki tarah kaam karta hai jo ek bade house (physical server) mein multiple rooms (VMs) ko manage karta hai. Har room ko space aur resources dekar, ek hi house mein kai log (virtual systems) rehsakte hain, jo space aur power bachata hai aur IT management ko asaan banata hai.

- ESXI is Hypervisor
- ESXI take physical resource from physical server and assigned to VM
- Assigned resources from physical server to VM
- Reduced data center space , power , IT administration services

ESXI


ESXI
App
OS
VM
Hypervisor(ESXI)
Physical Server
CPU
Memory
Network Card
HD
Physical Resources

ESXi
Installation Steps



ESXi
Installation Steps

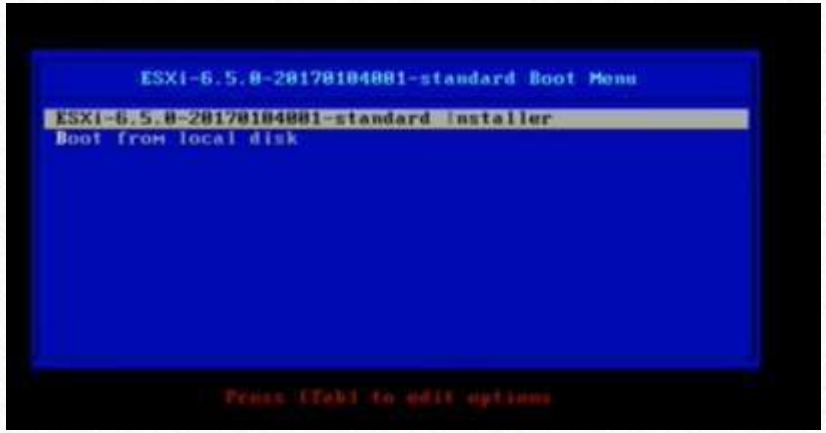
- **Topics to be Covered**:

  What is ESXI

  Lab: Installation steps

Installation Steps

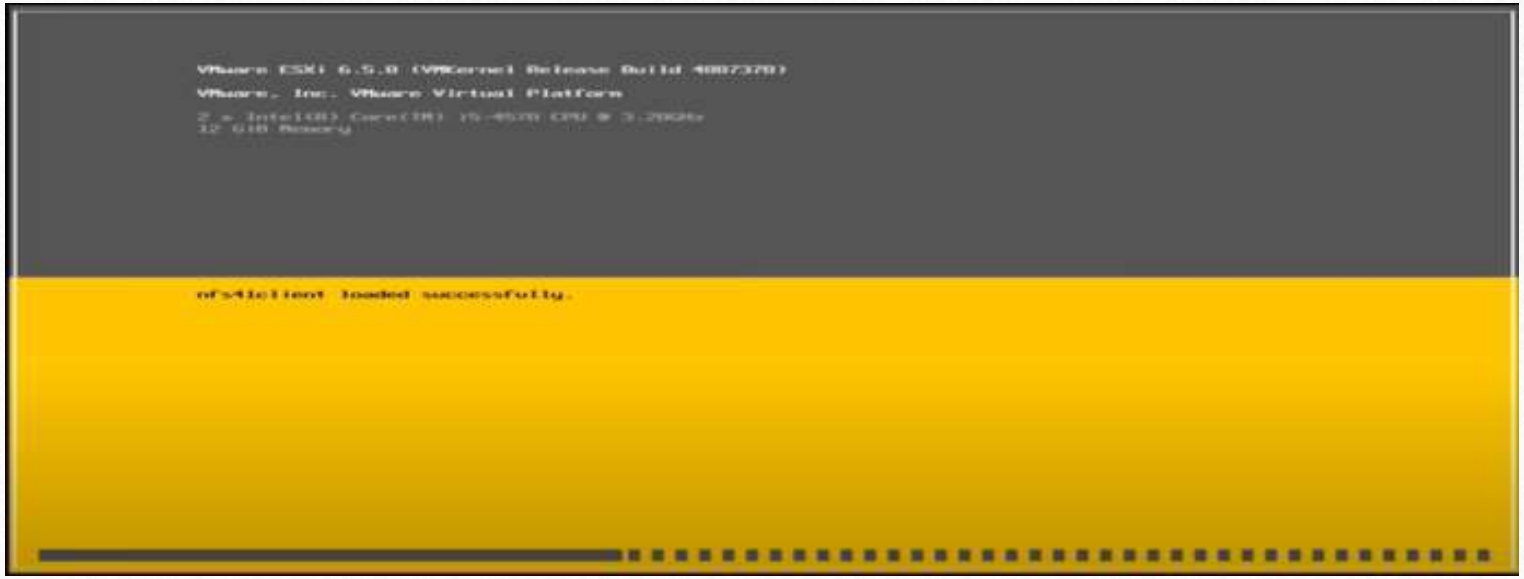
- Download **ESXi 6.5 Image from VM ware website**
- Boot System with Bootable ESXI Image



Installation Steps

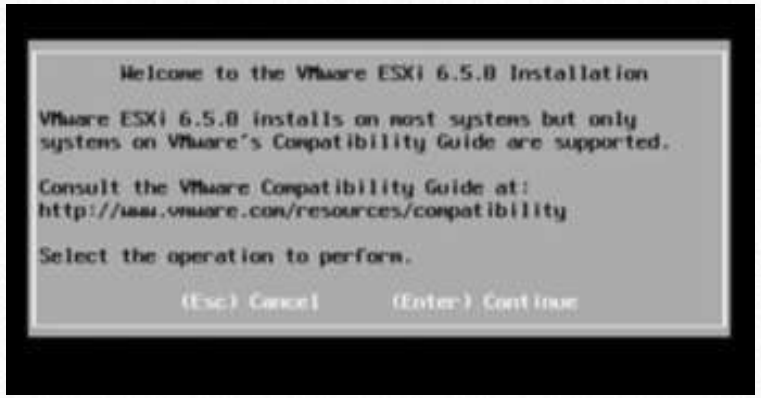


Installation Steps



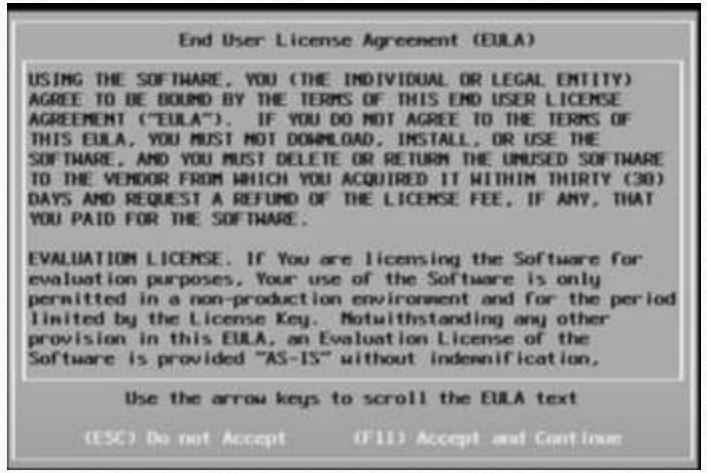
- This is not new one

Installation Steps



- Find HW and then used that compatible image
- Enter continue

Installation Steps



Installation Steps



- Scan disk to install ESXI
- Single disk used by ESXI

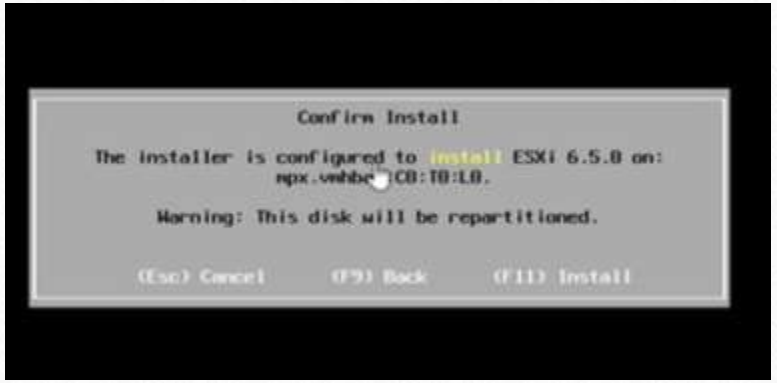
Installation Steps



Installation Steps



Installation Steps



Installation Steps



Installation Steps



Installation Steps

- Remove IOS image



ESXi
Configuration Steps

ESXi
Installation Steps

- **Topics to be Covered**:

  What is ESXI

  Lab: Installation steps

  Lab: 2 Configuration Steps

Configuration Steps



Configuration Steps



- As authenticate ESXI Host
- Enter user name and pass

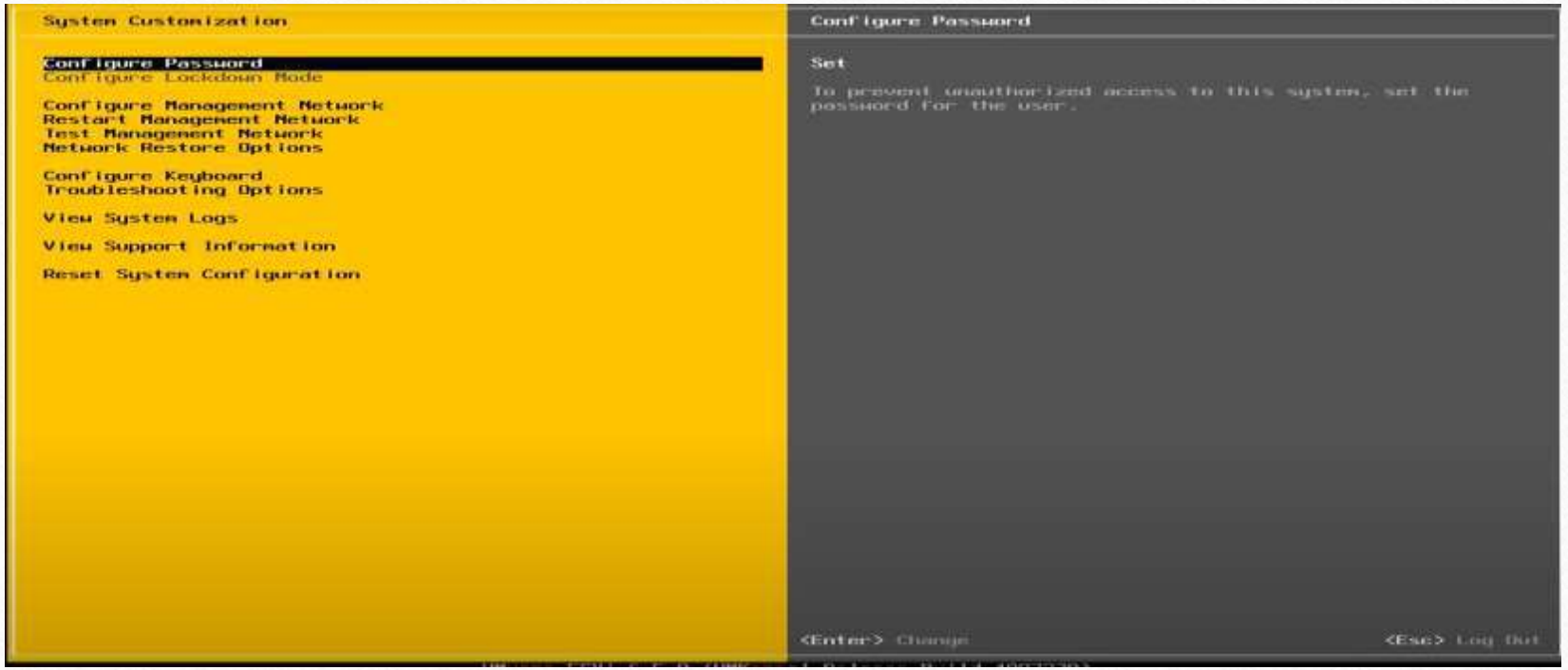
Configuration Steps (System Customization)



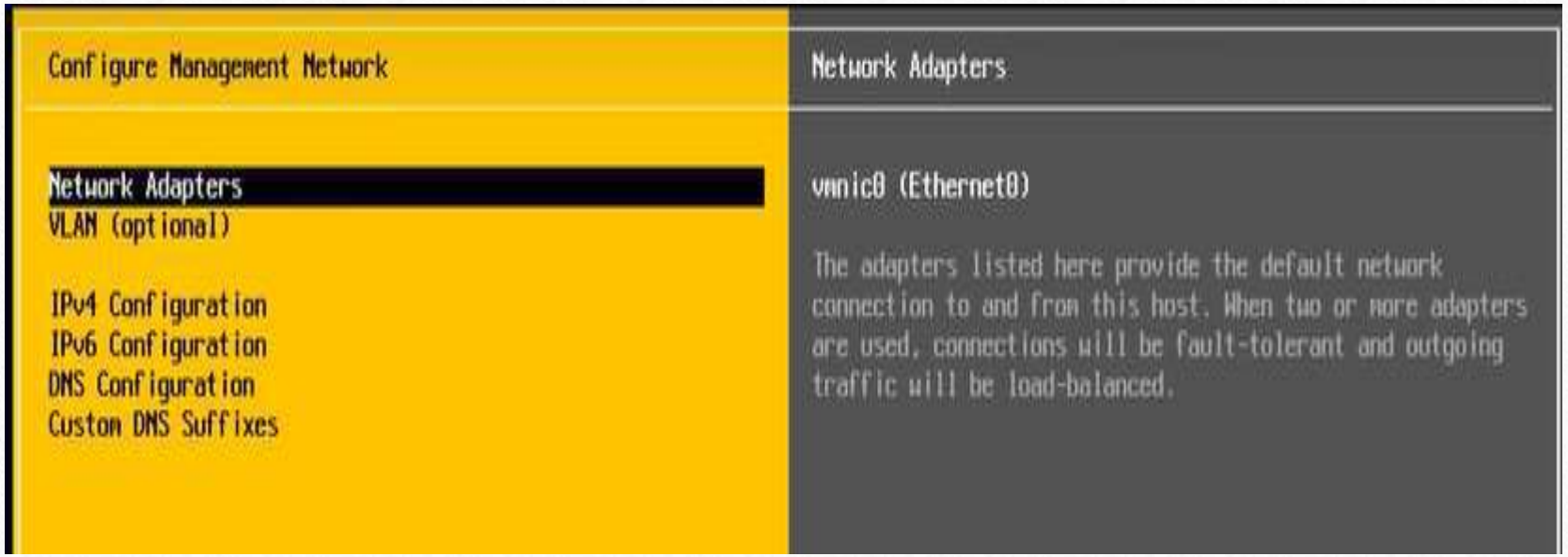
Configuration Steps
Password



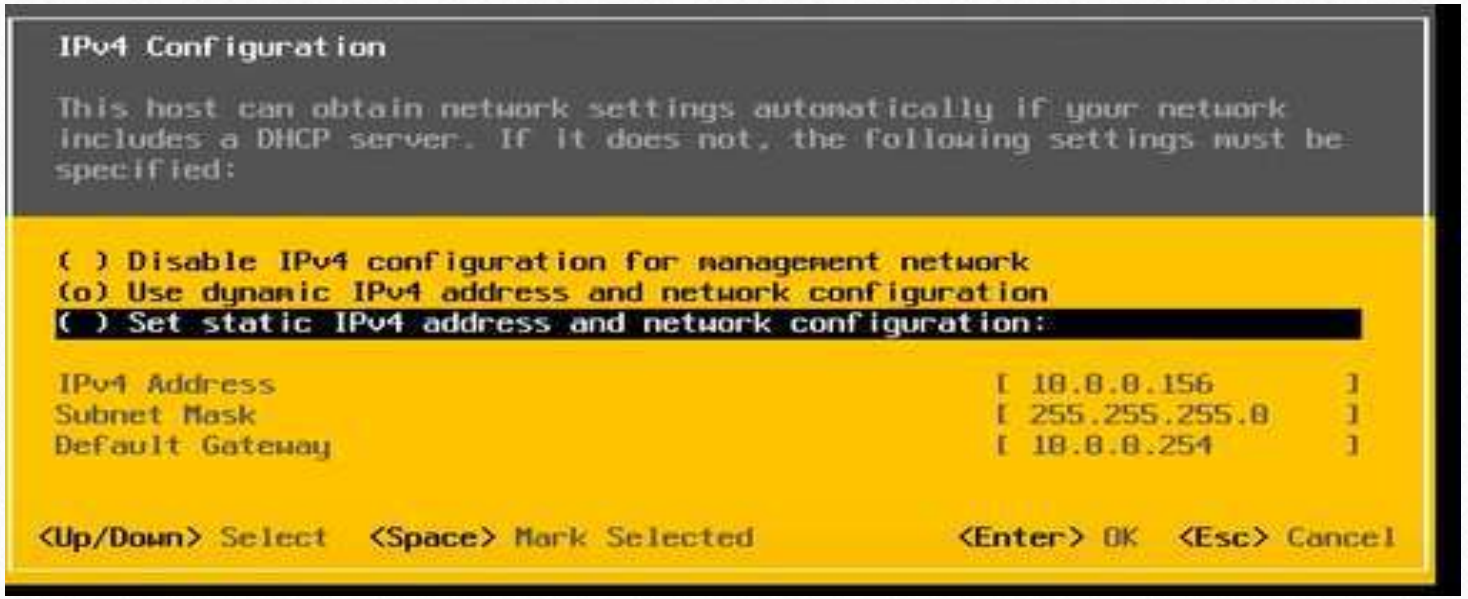
Configuration Steps
Configure lockdown Mode



Configure Management Network



Configure Management Network



Restart Management Network

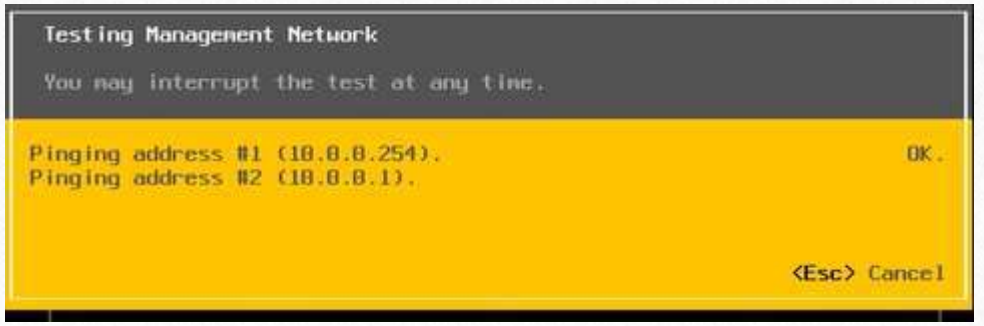


- If you have problem to connecting ESXI Host you can restart the management network

Test Management Network



Test Management Network

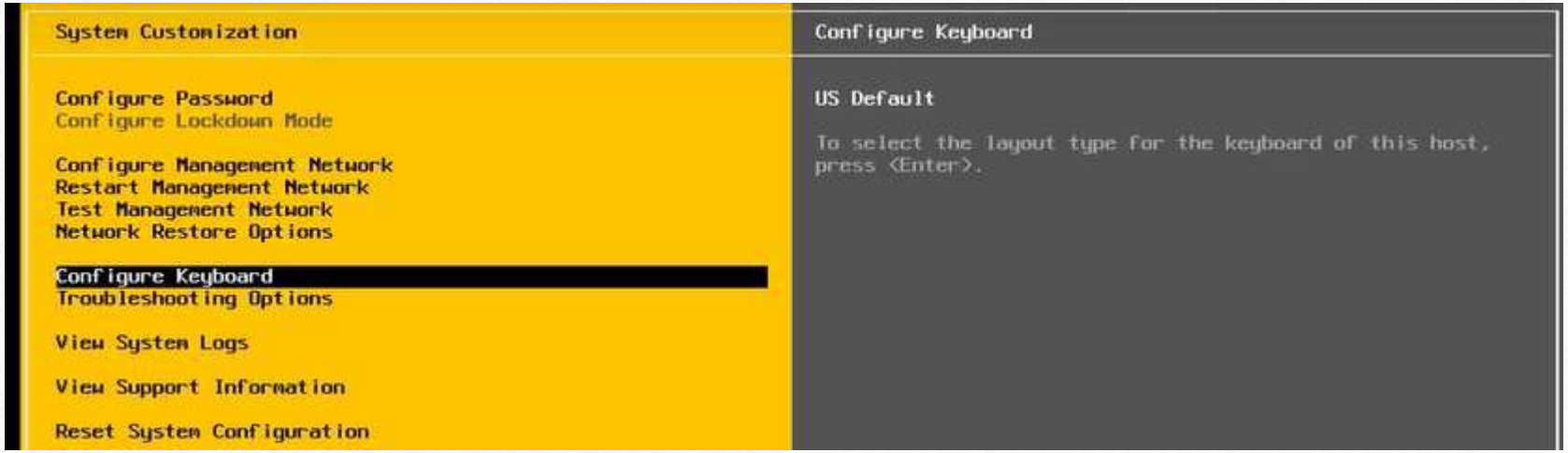


- Ping getaway

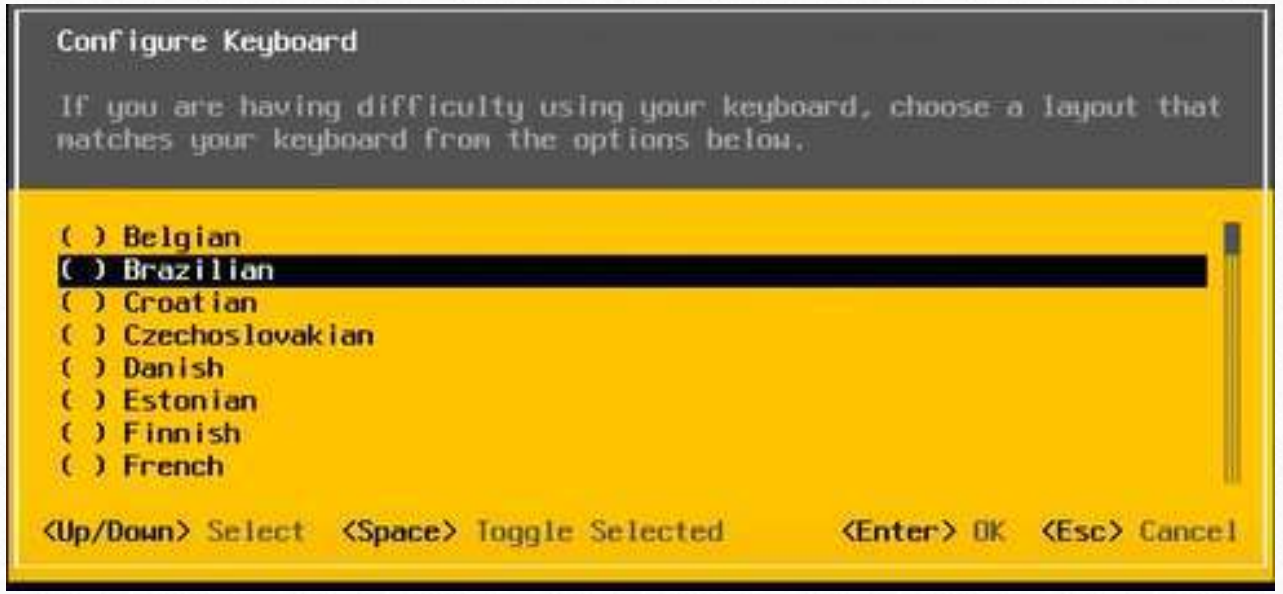
Network Restore Options



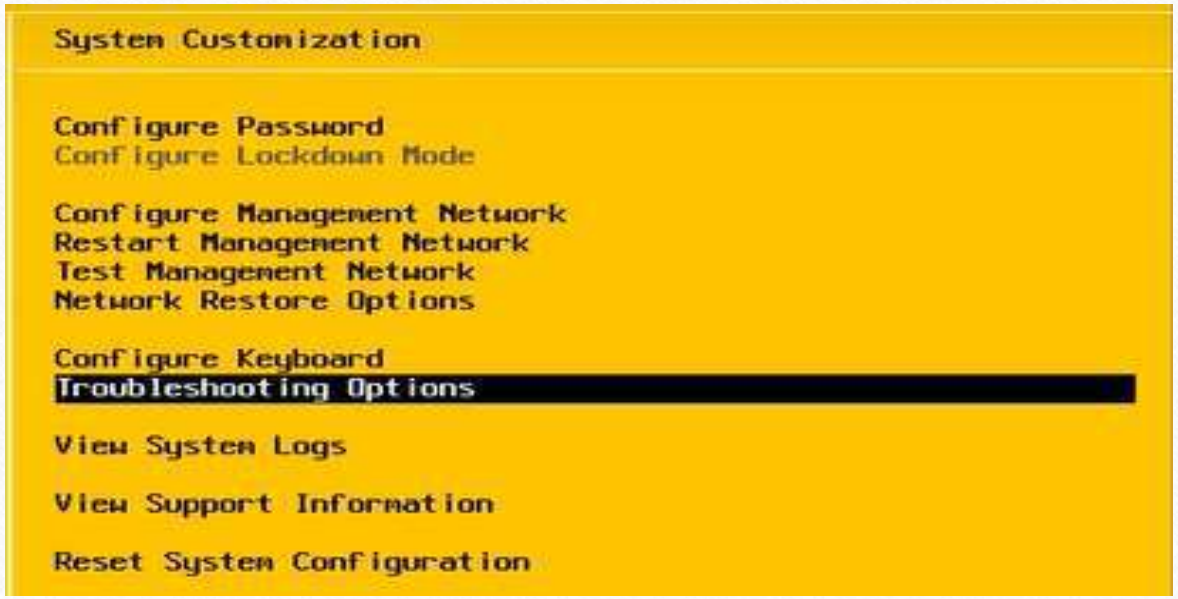
Configure Keyboard



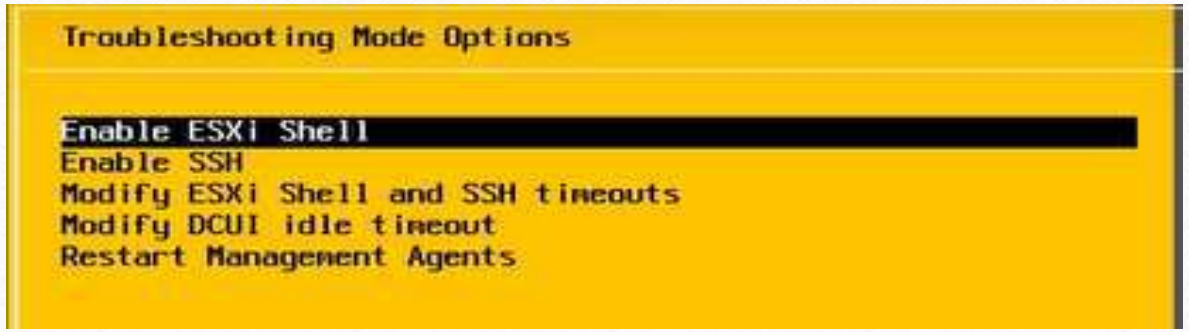
Configure Keyboard



Trouble shooting options



Trouble shooting options



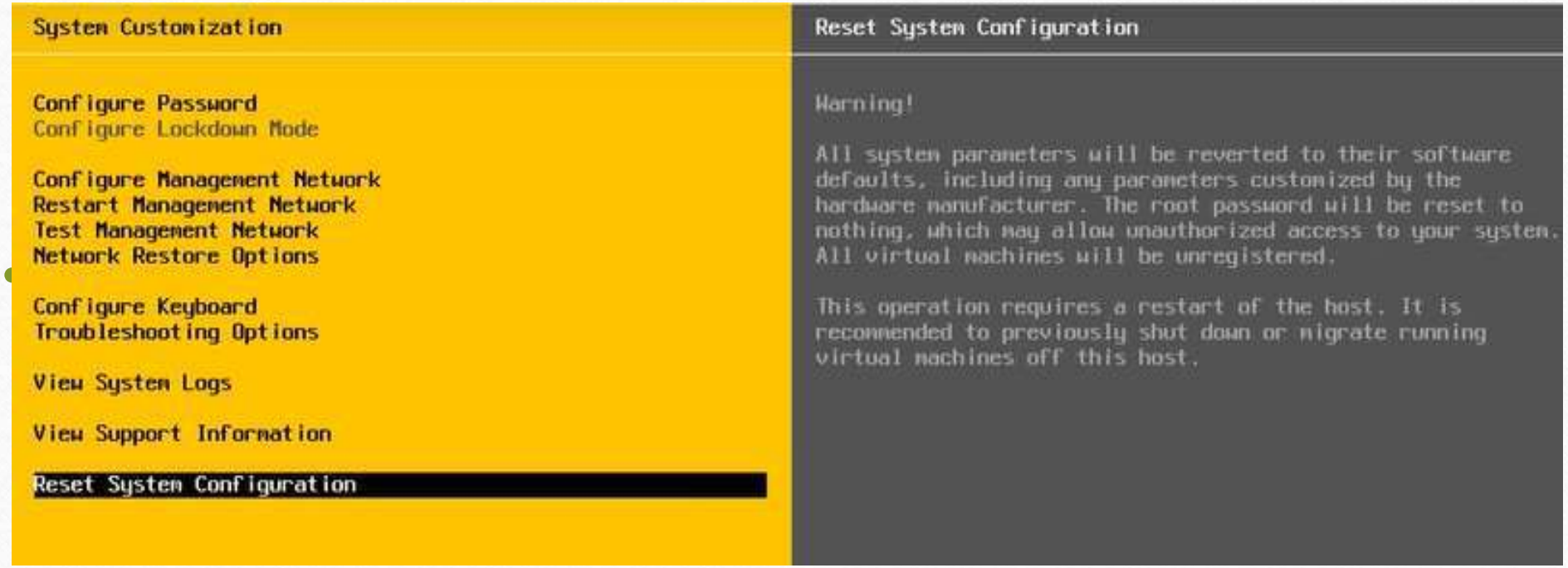
View System Log



View Support Information



Reset System Configuration



ESXi
Creating Virtual Machine



ESXi
Creating Virtual Machine

- **Topics to be Covered**:

  What is ESXI

  Lab: Installation steps

  Lab: 2 Configuration Steps

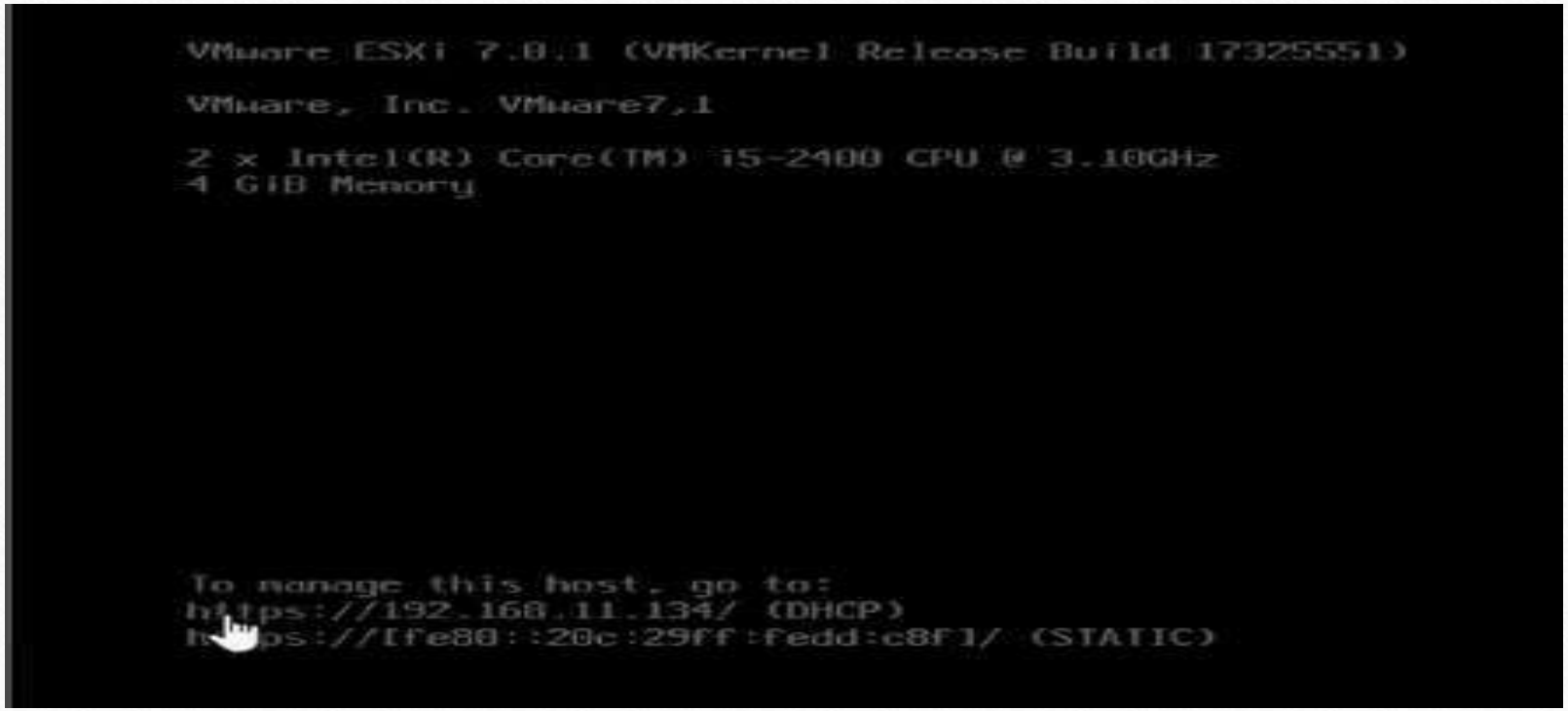
  Lab 3: Creating Virtual Machine

Recommended Steps

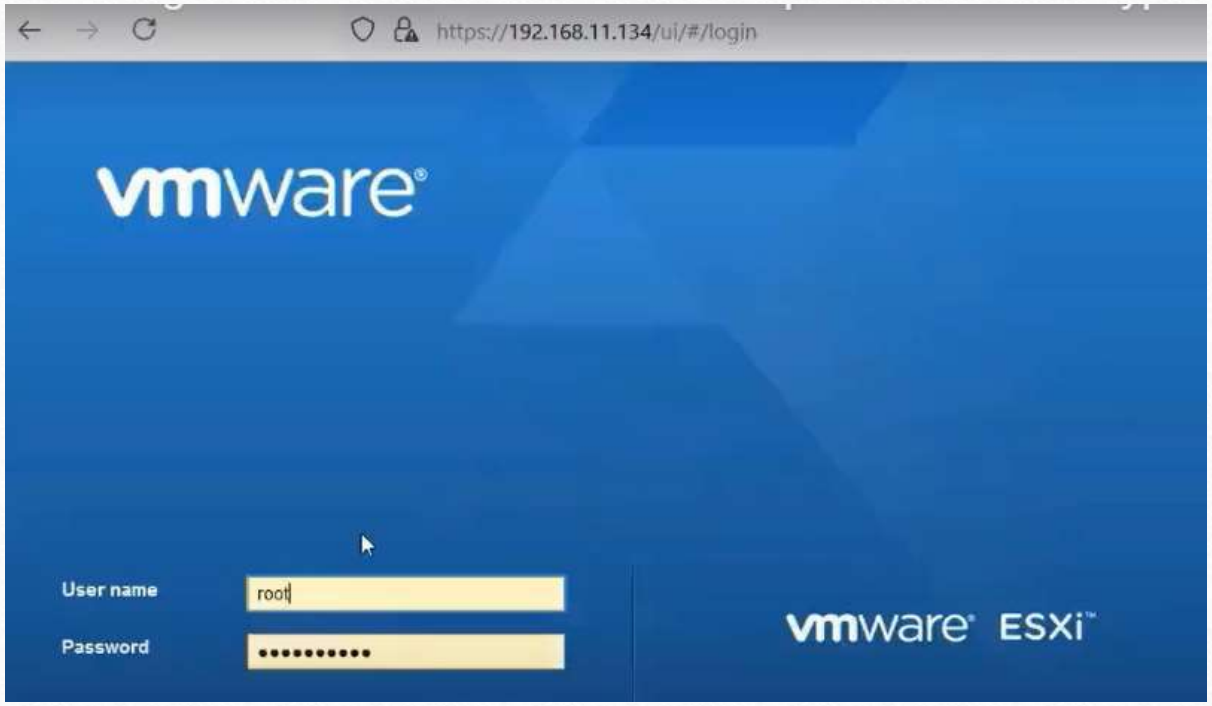


- Creating Data Store
- Upload image of OS
- Installing VM

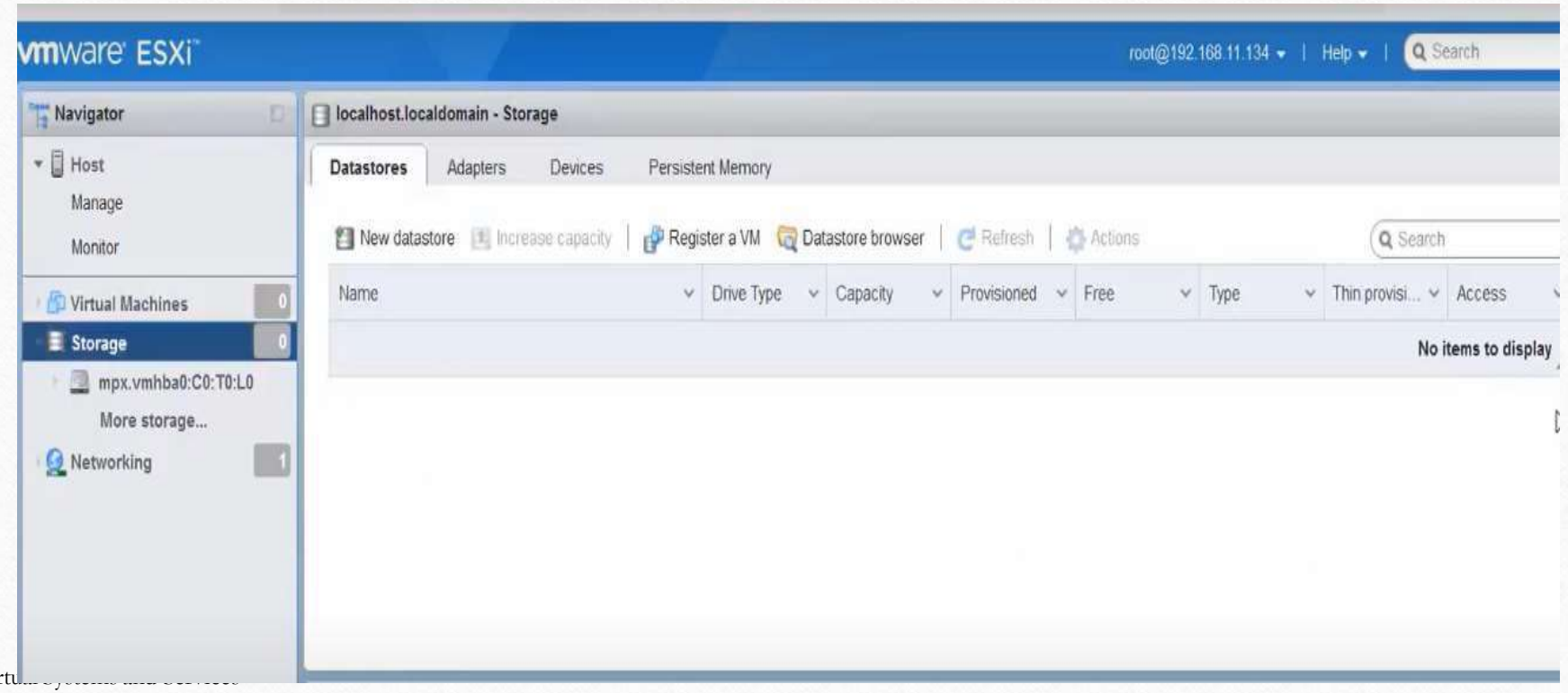
Creating Data Store



Creating Data Store



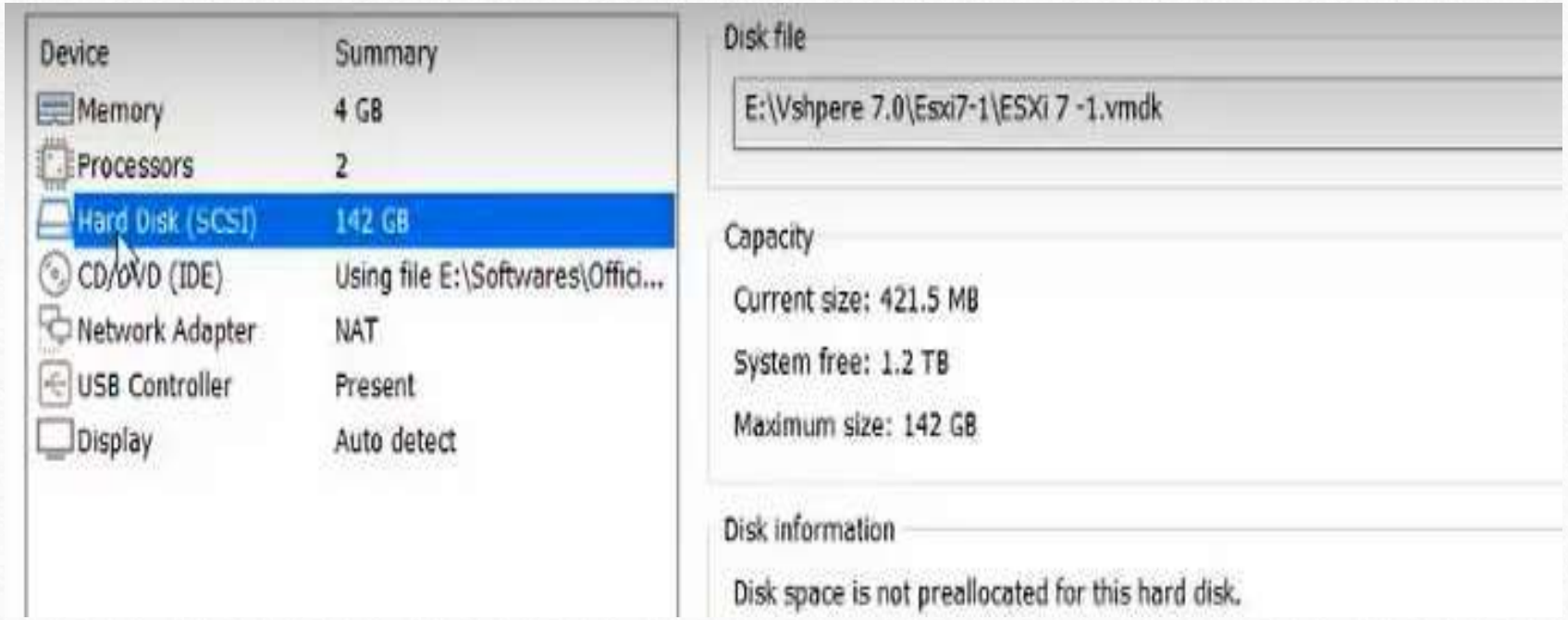
Creating Data Store



Creating Data Store



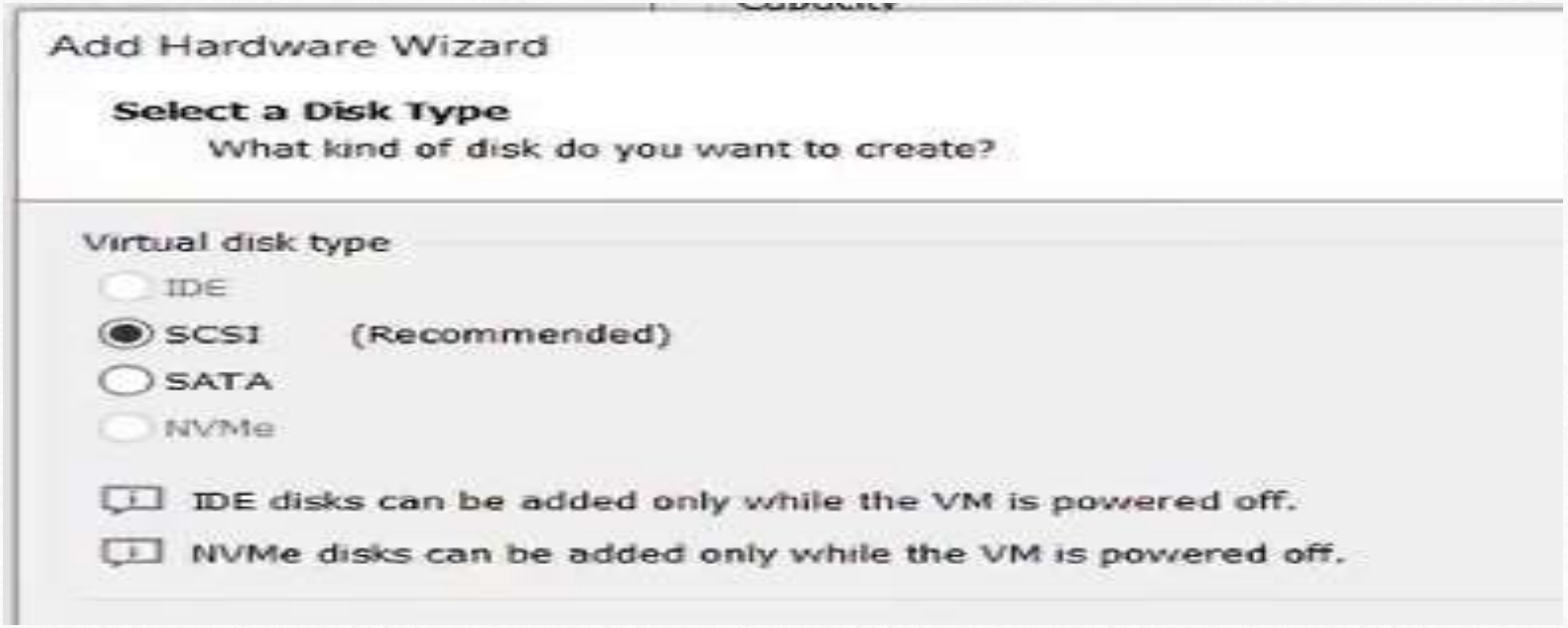
Creating Data Store



Creating Data Store



Creating Data Store



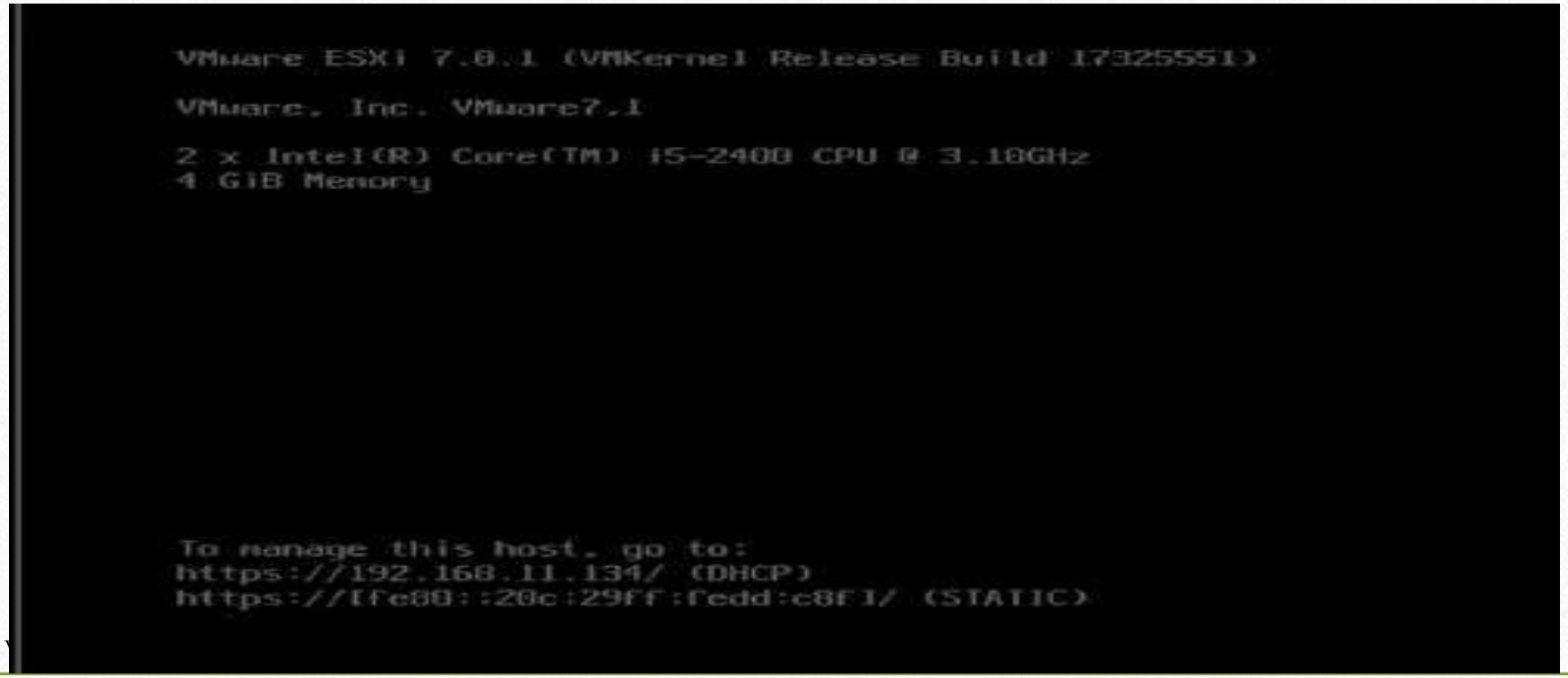
Creating Data Store



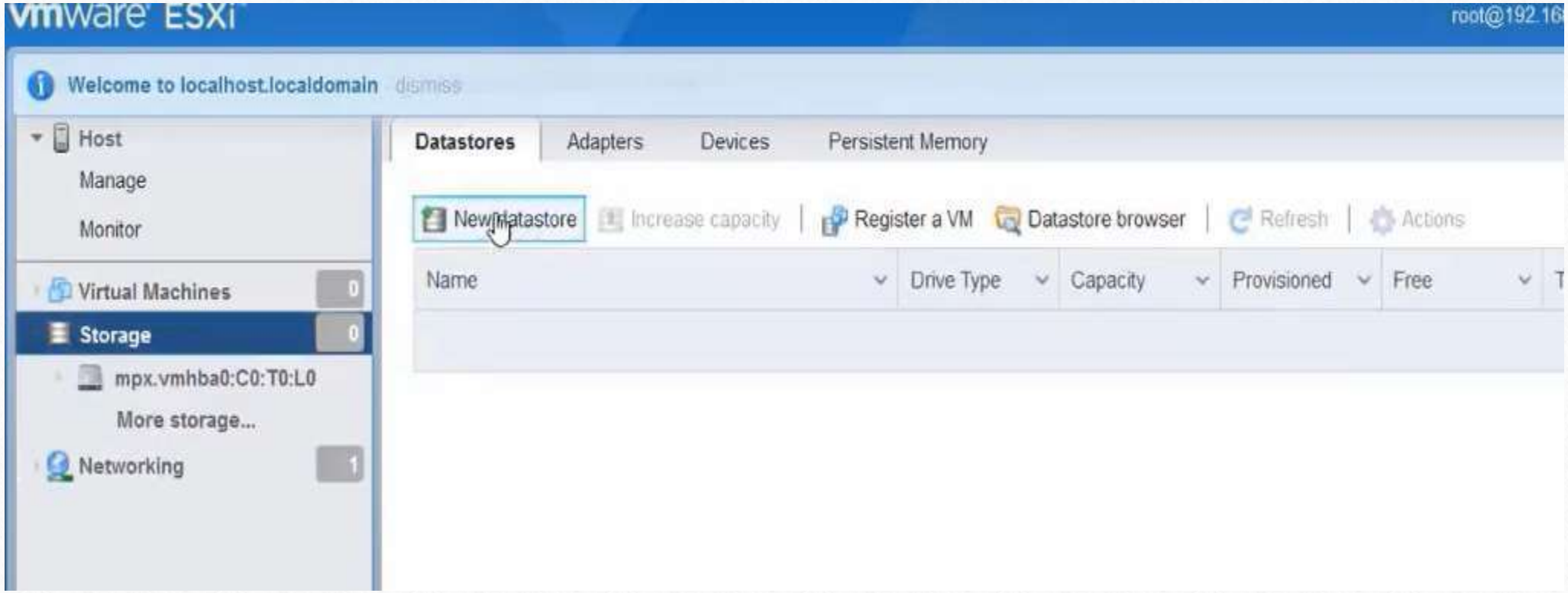
Creating Data Store



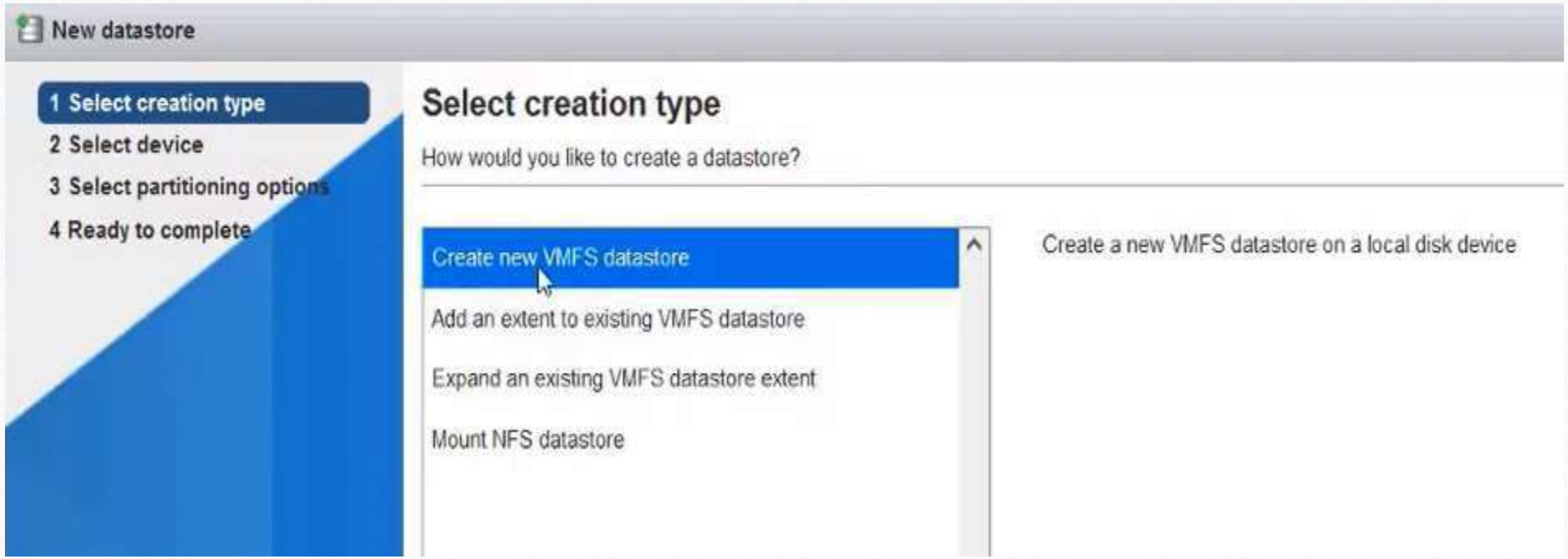
Creating Data Store
Restart



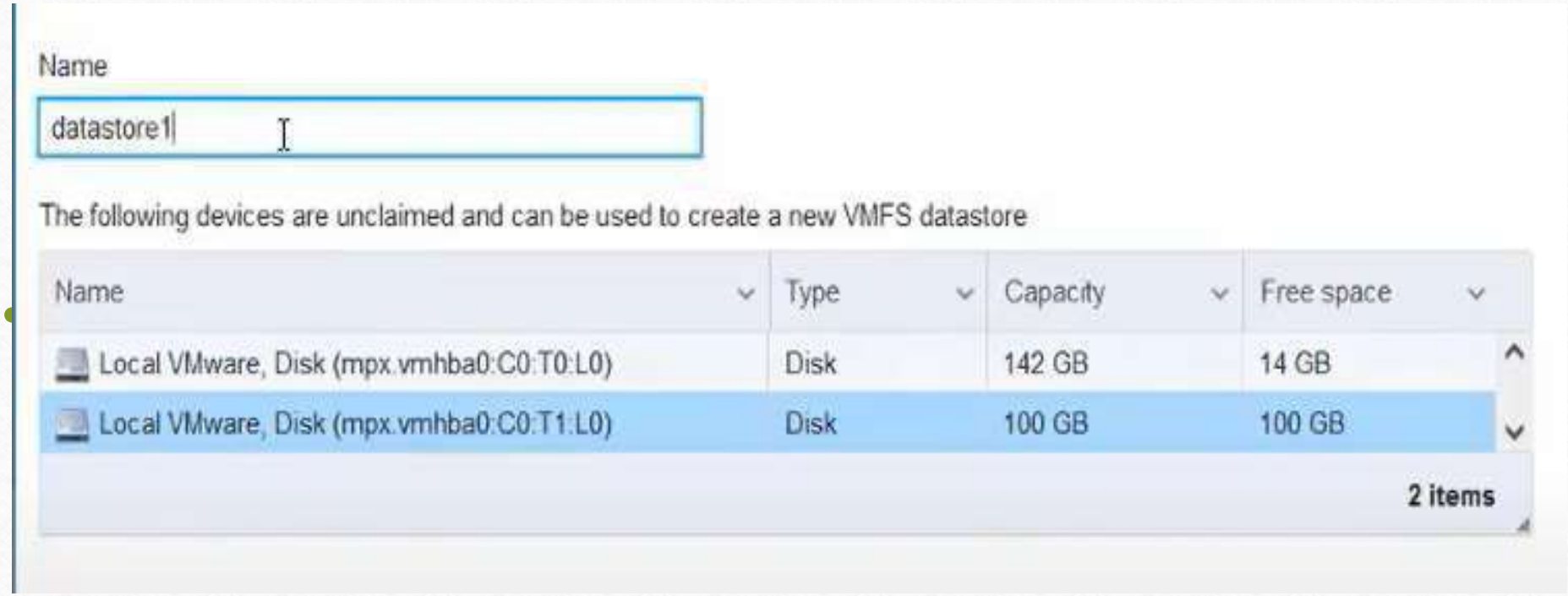
Creating Data Store



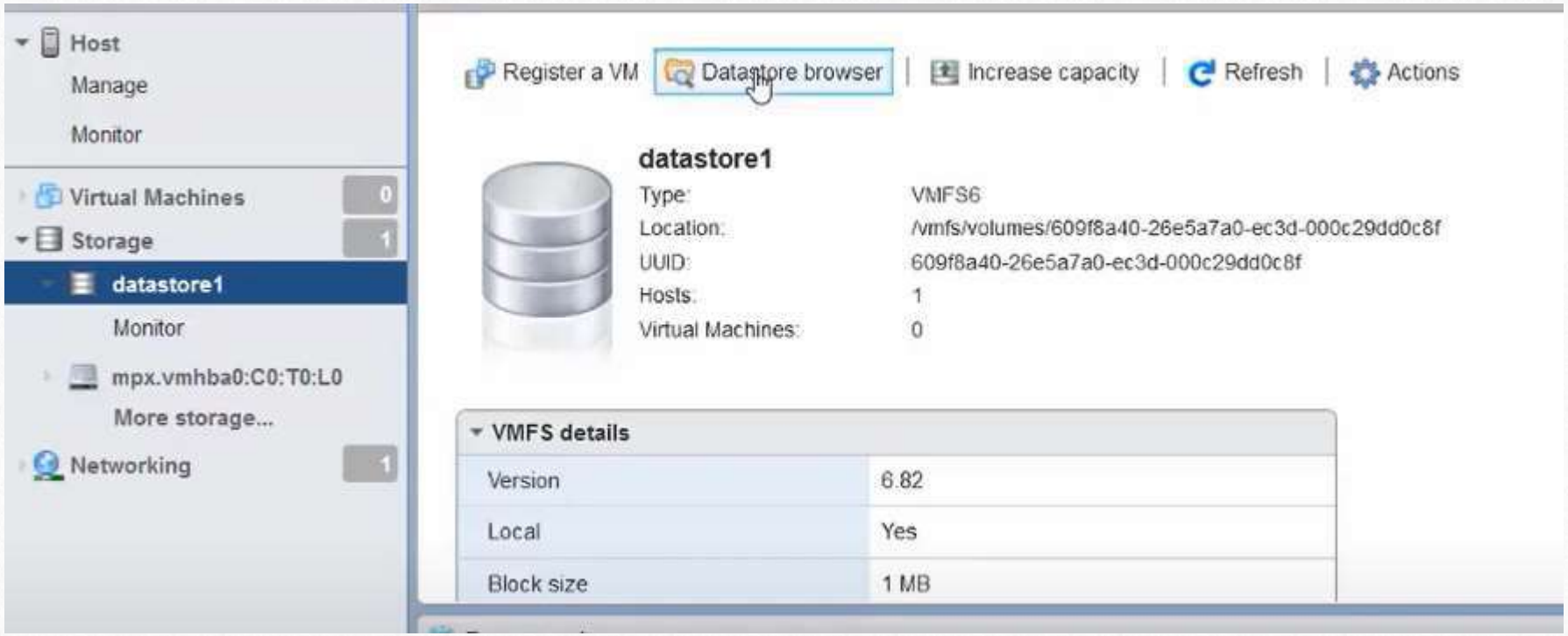
Creating Data Store



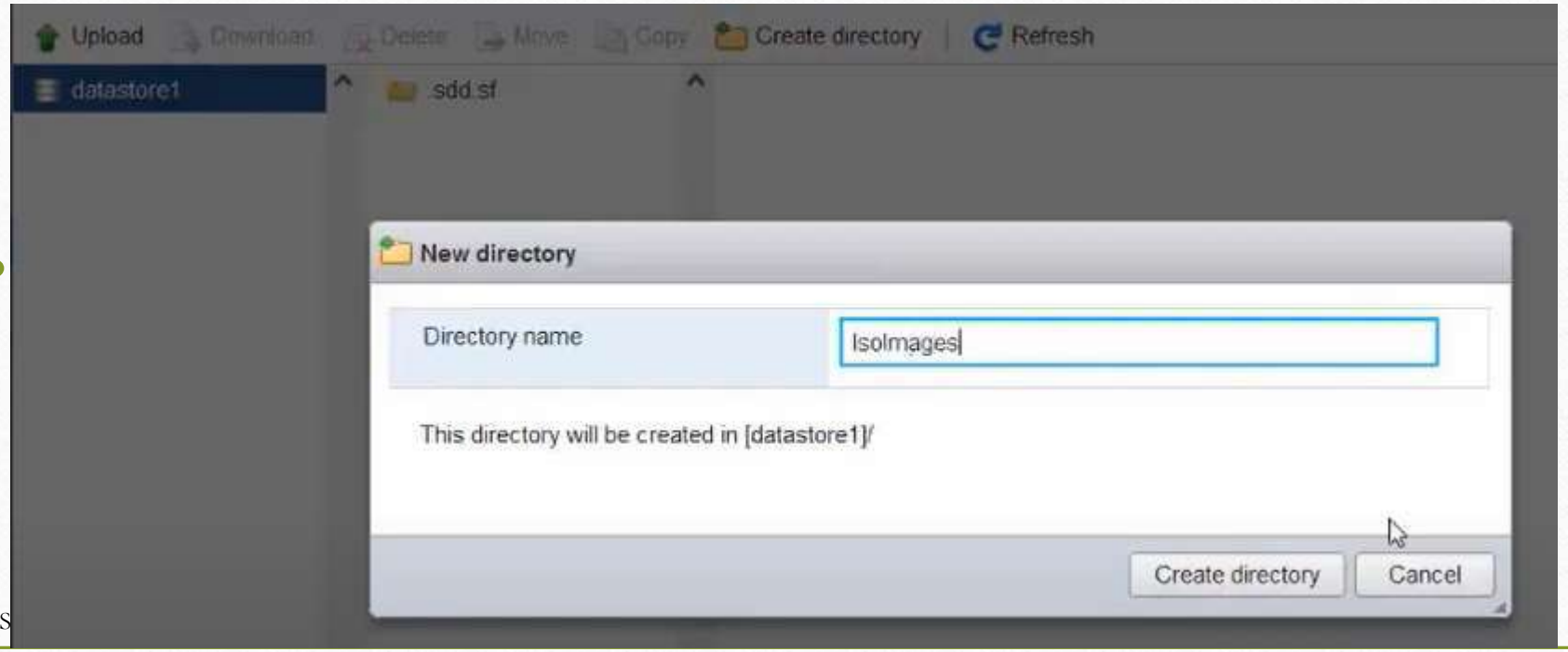
Creating Data Store



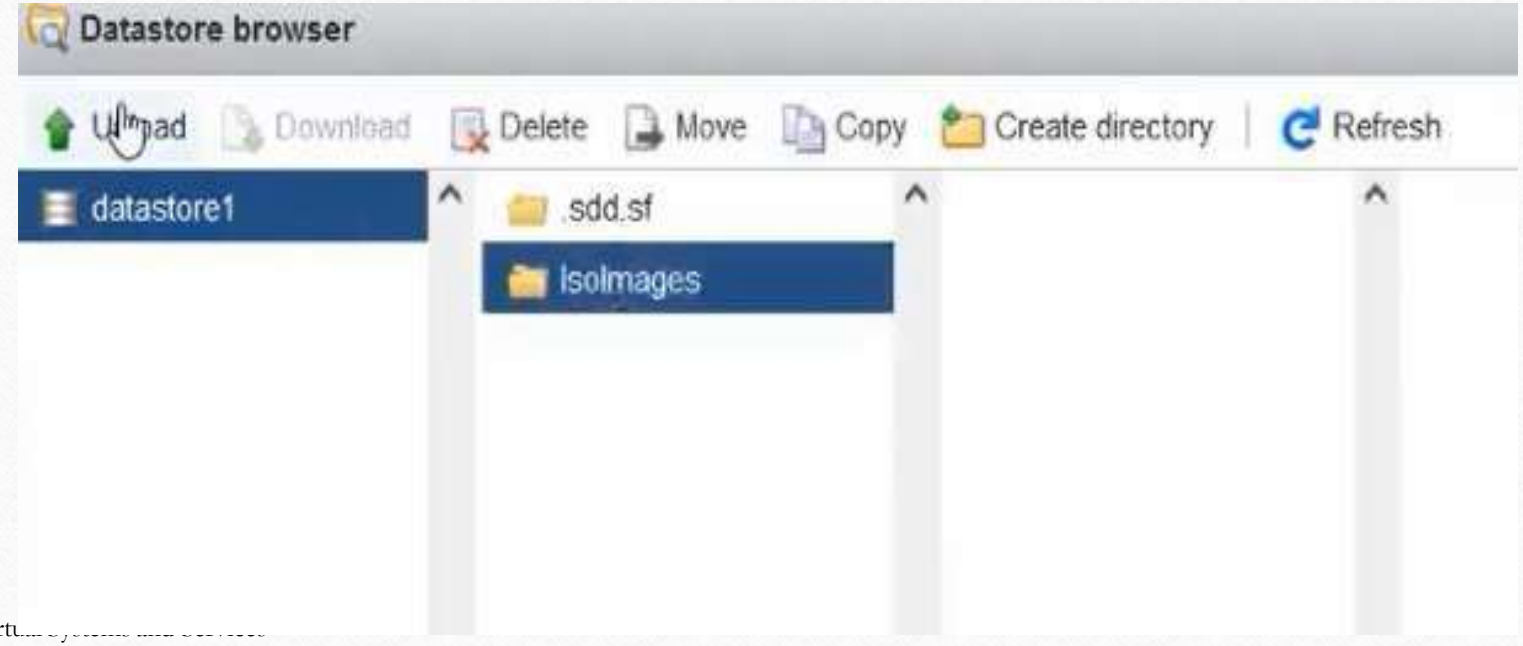
Create Directory on Data Store



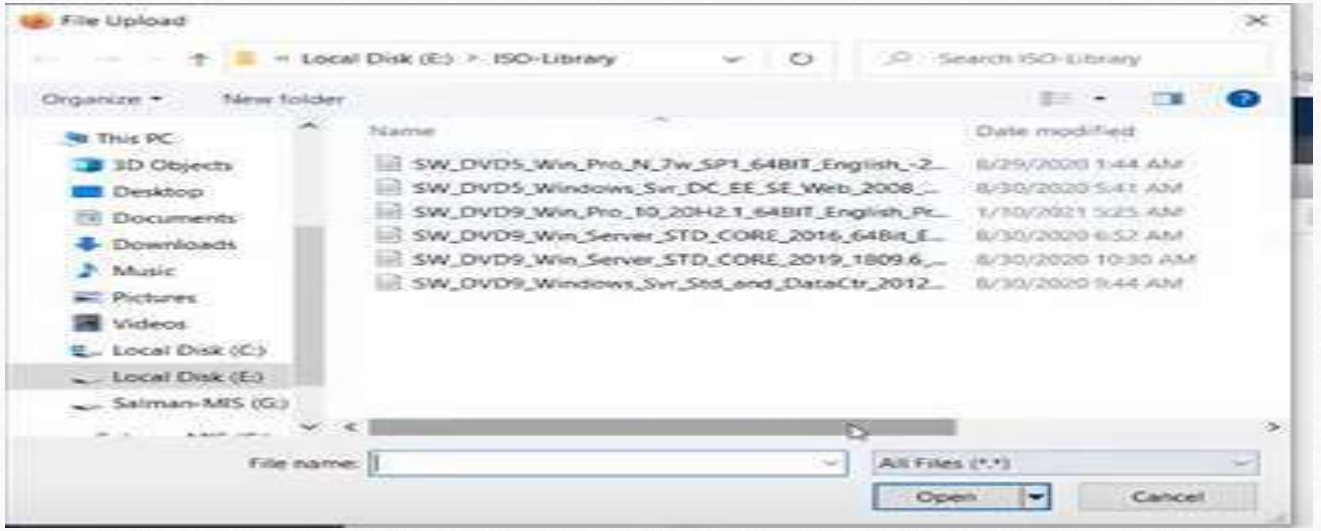
Create Directory on Data Store



Upload OS image



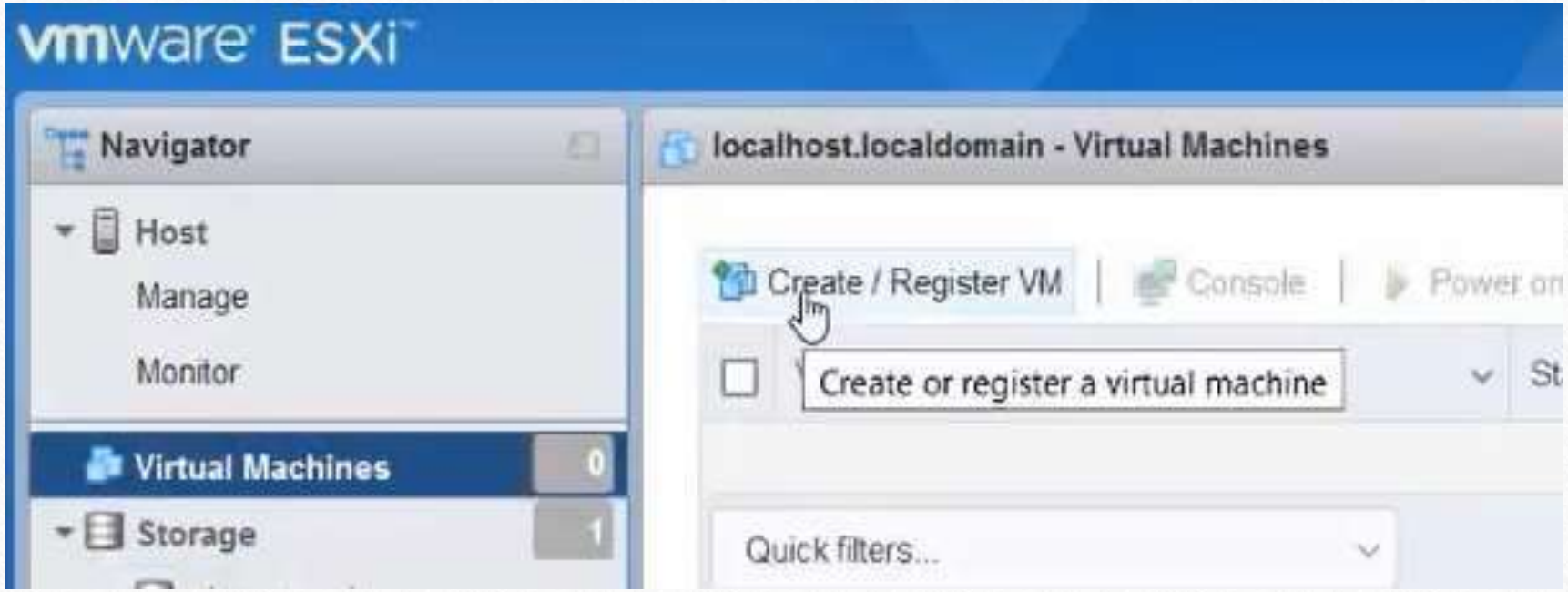
Upload OS image



Upload OS image



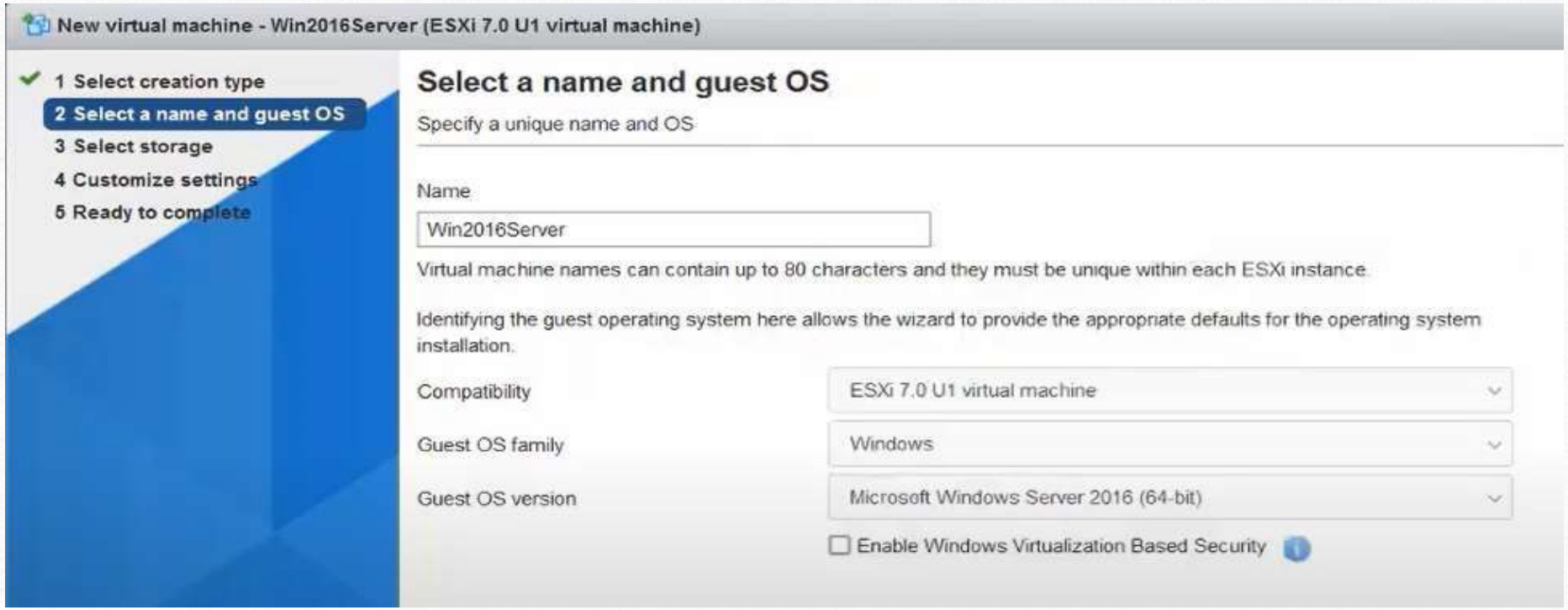
Installing VM



Installing VM



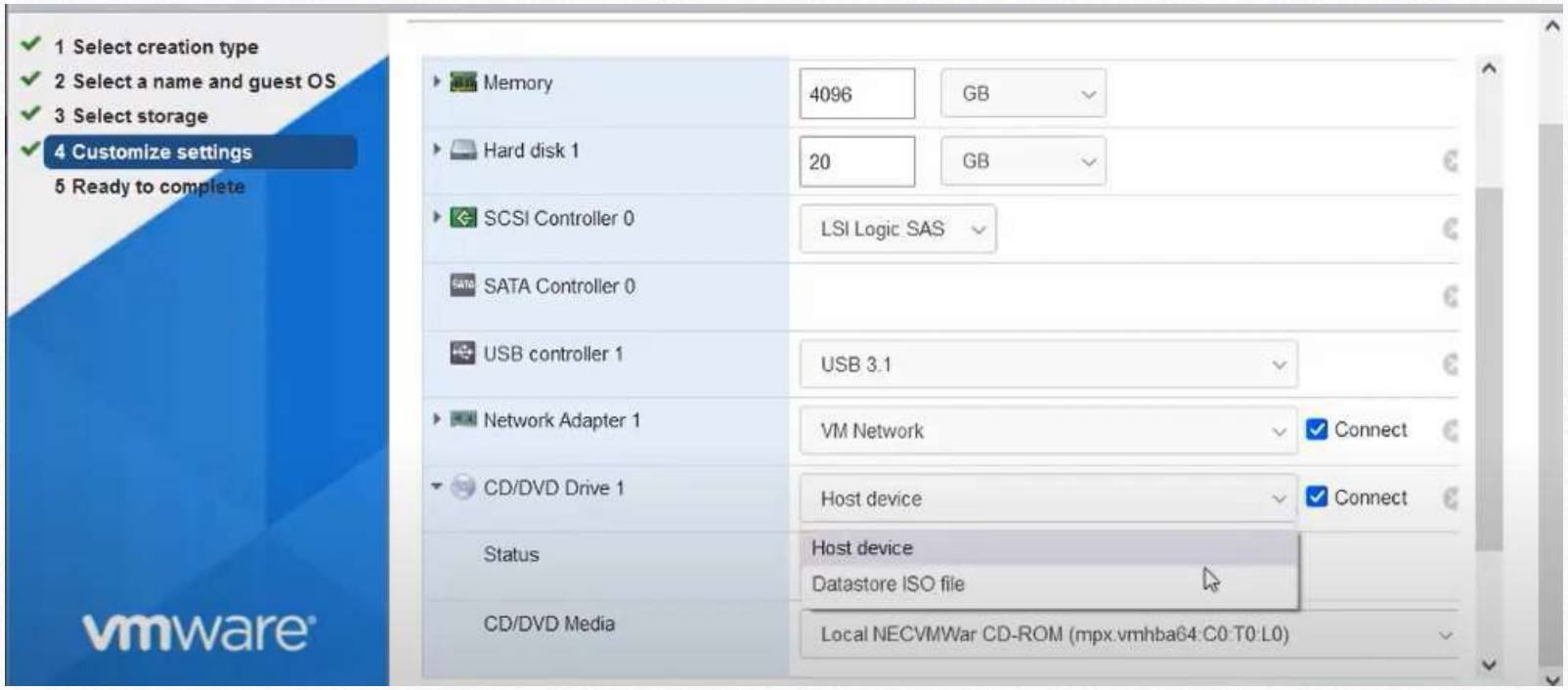
Installing VM



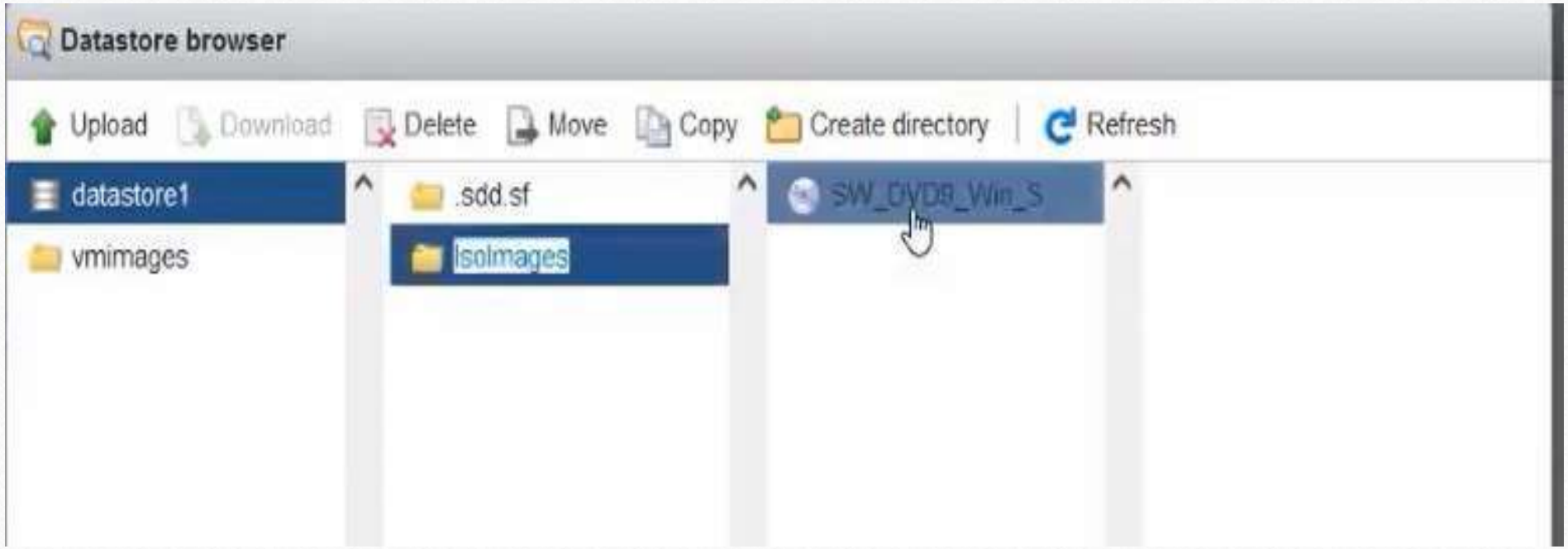
Installing VM



Installing VM



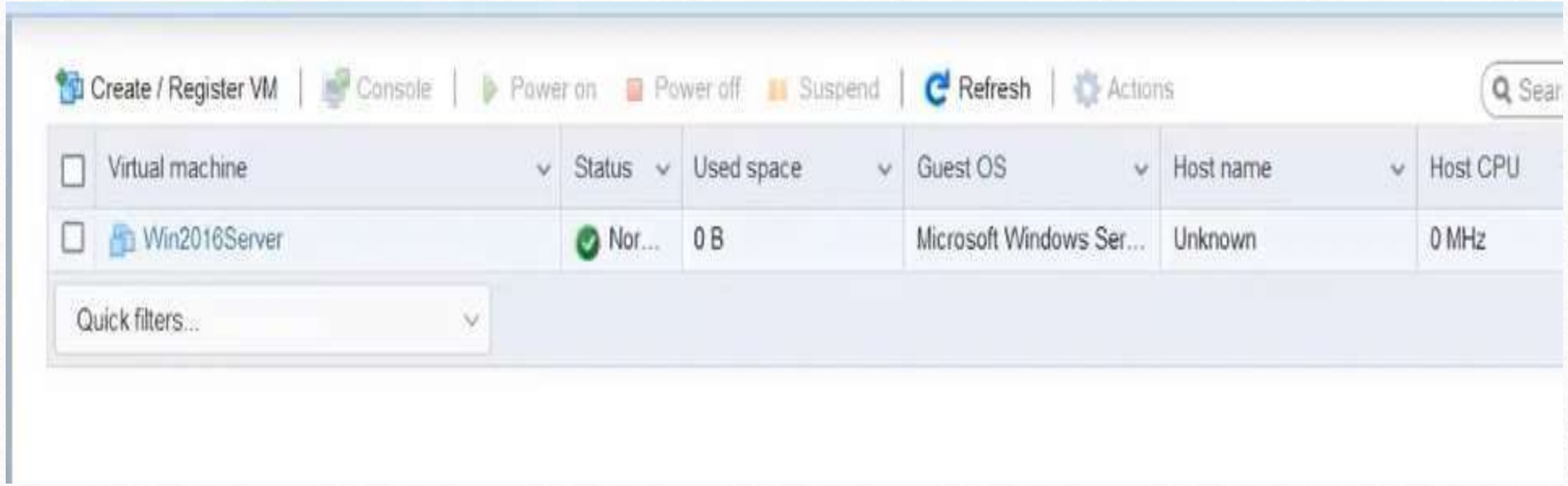
Installing VM



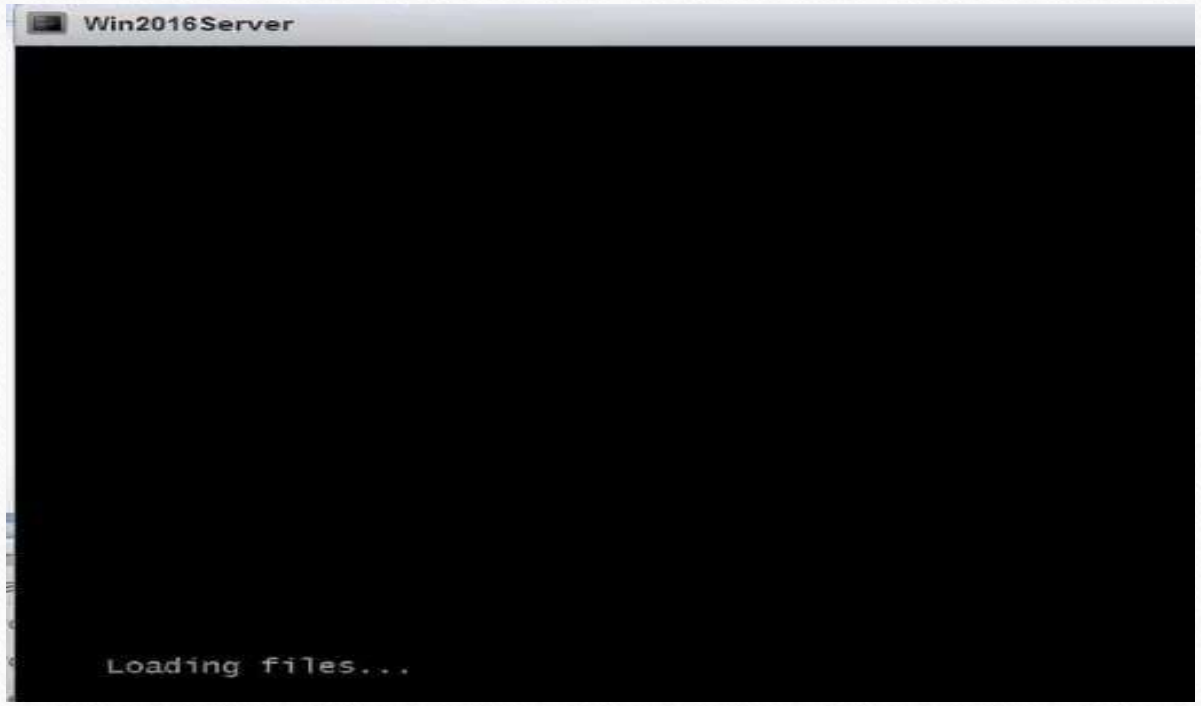
Installing VM



Install Window



Install Window





VirtualBox



Virtual Systems and Services

VirtualBox



Work on multiple platforms

- Oracle VM VirtualBox is a cross-platform virtualization product
- It is used for virtualizing x86 and AMD64/Intel64 computing architecture
- It is freely available as open source software under the terms of GNU General public license (GPL)
- It runs on all major operating systems including Windows, Linux, Mac and Solaris

VirtualBox

- Different Versions of Oracle VM VirtualBox exists to install (for Windows, Linux and Solaris etc.)
- VM VirtualBox is first downloaded and installed on any host OS
- It has both the graphical and Command Line interface (GUI and CLI)
- Using VirtualBox, multiple guest operating systems can be installed as VMs on host OS
- We can install Windows 7 or Linux as guest OS on hosted OS Windows 10

VirtualBox



- VirtualBox supports both the hardware and software virtualization
- It is recommended to use VirtualBox for small and mid-range projects

VMware Vs VirtualBox



Virtual Systems and Services



VMware Vs VirtualBox



VMware

- It is a virtualization software platform developed by vendor, VMware
- It is a type-1 hypervisor
- VMware supports hardware virtualization
- Its workstation player is free while other VMware products require a paid license.

VirtualBox

- It is a virtualization software platform developed by Oracle Corporation
- It is a type-2 hypervisor
- VirtualBox supports both software & hardware virtualization
- It is an open source virtualization product

VMware Vs VirtualBox



VMware

- VMware workstation works on Windows and Linux. For macOS, we need to install VMWare Fusion/Fusion Pro
- It supports VMDK disk format

VirtualBox

- VirtualBox can install on Windows, Linux, Solaris, MacOS, FreeBSD
- It supports VDI, VHD, VMDK disk formats

Virtual Box Disk Image, Virtual Hard Disk, Virtual Machine Disk

VMware Vs VirtualBox



VMware

- VMware supports snapshots on its paid products
- VMware Virtual Machines run faster
- For enterprise projects, VMware is recommended

VirtualBox

- VirtualBox supports snapshots free of cost
- VirtualBox Virtual Machines run slower comparatively.
- For small and mid-level projects, VirtualBox is recommended



Hyper-V



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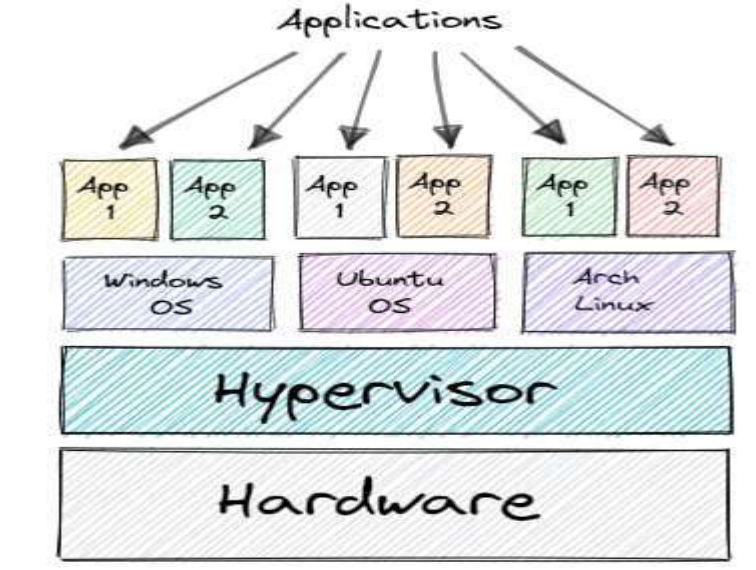
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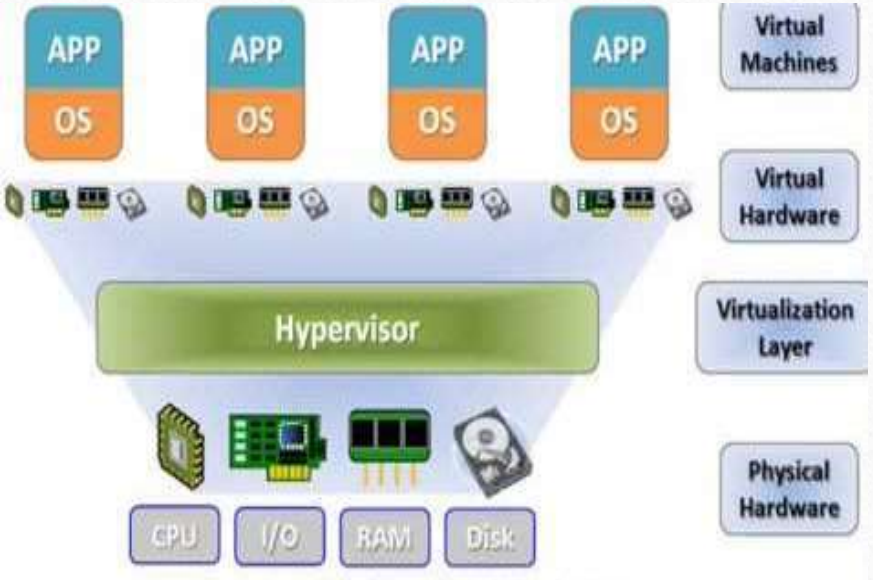
Hyper-V

- Hyper-V is a virtualization platform through which multiple isolated environments can be created on single physical server. like health care system
- Hyper-V is the product of Microsoft
- Hyper-V can be enabled from the Windows operating system
- Hyper-V is type 1 hypervisor

Type 1 Hypervisor

- Hyper-V is type 1 hypervisor





Advantages of Hyper-V

- Cost can be reduced
- Hardware can be utilized with full capacity
- Provide flexibility
- Different environments like testing and production environment can be created
- Power consumption can be reduced too

Compare

- Hyper-V
  - Microsoft dependent
  - Checkpoints
  - Hyper-V live migration tool for migrating workload
  - Hyper-V provide support for Linux
- VMware
  - Complete echo system for virtualization
  - Snapshots
  - VMware vMotion for migration of work load
  - Fault tolerance and disaster recovery

Conclusion

- Hyper-V is Virtualization platform
- All advantages of virtualization can be achieved with the help of Hyper-V
- Both VMware and Hyper-V are the market players of Virtualization
- Hyper-V is type 1 hypervisor

Image sources



- https://www.youtube.com/watch?v=tlL1W9MLobo
- https://dev.to/kcdchennai/virtual-machines-101-2gmg
- http://www.greenit-monaco.com/en/software/hyper-v.html

Hyper-V Architecture



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Hyper-V Architecture

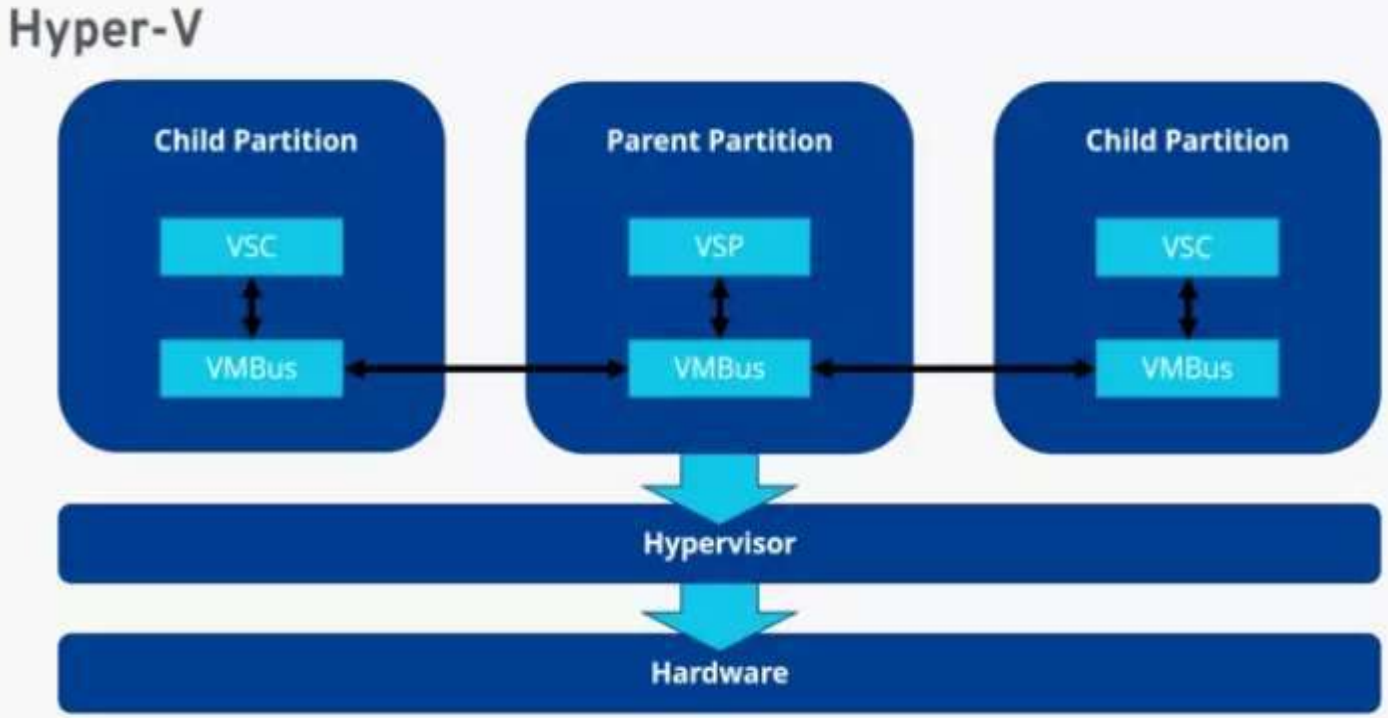


Image Source:https://www.ionos.com/digitalguide/server/know-how/what-is-hyper-v/

Hyper-V Lab

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Hyper-V Lab

- Hyper-V is a Microsoft Product for Virtualization
- Hardware Virtualization
- Flexibility
- Modify your infrastructure according to your workload

Step 1

- Enable Hyper-V in Windows 10 pro or Windows Server
- Go to Apps & features
  - Then Programs & Features
  - Turn Windows features on or off
  - Check Hyper-V option from the list so that it needs to be selected .
- Now Hyper-V will be accessible from the search bar in windows

Step 2

- Create a Virtual Machine
  - Search Hyper-V Manager open it
  - From Hyper-V Manager Dashboard in Actions Click on New
  - Click on Create Virtual Machine
  - Configure it according to your needs

Step 3



- Install Operating system in the Virtual Machine
- Select Virtual Machine you just created in step 2 which will be shown in the center
- From the menu bar Click on Actions and then on Connect then proceed with Start option on Screen
- Now your Operating System installation step will start

Step 4



- Play with Virtual Machine state
- Either turn it off if its running
- Or turn it on if its off
- Also check Shutdown status
- Right click on the Virtual Machine and Look for above Options

Step 5



- Explore the setting option of the Virtual Machine
- Increase the memory or CPU according to your workload from the setting option of VM

Try it Yourself



- Now its your turn explore and apply all options covered in this Lab

Merits and Demerits of Virtual Machine

Merits of Virtual Machine

- **Virtualization**
  - Virtual Machine transforms traditional hardware into virtual environment
- **Run Multiple Operating Systems**
  - VMs allow us to run multiple operating systems on a single host
- **Reducing cost:**
  - Several VMs can host on a single physical server, reducing the cost of data center

Merits

- **Portability**
  - VM can be easily migrated across the Virtual Environment
- **Security**: VMs run in isolation. If security is compromised in one VM, then no effect on other VMs
- Runs incompatible Software
- If one VM fails, then other machines will not be affected
- Virtual Machine enables administrators to simplify backup, disaster recovery, new deployment

Demerits of Virtual Machine

- **Cost**
  - Running VMs with cloud computing is expensive
- **Low Performance**
  - VM performance is less than the real machine
- **Less Efficiency**
  - Less efficient than real machine
- **Single point dependency**
  - If physical host down, then all the hosted VMs becomes useless

- Performance: runs independently but still relies on resources of host machine that are shared among the VMs
- Efficiency: VM access hardware indirectly, will make a request for hardware access
- Reducing cost: reduce data center cost by reducing cost of physical resources, space and power
- Security: VMs run in isolation from one another. If security is compromised in one VM, then no effect on other VMs.
- Run incompatible software: OS and applications are decouple from the underlying OS

Virtual Machine



Topics to be covered



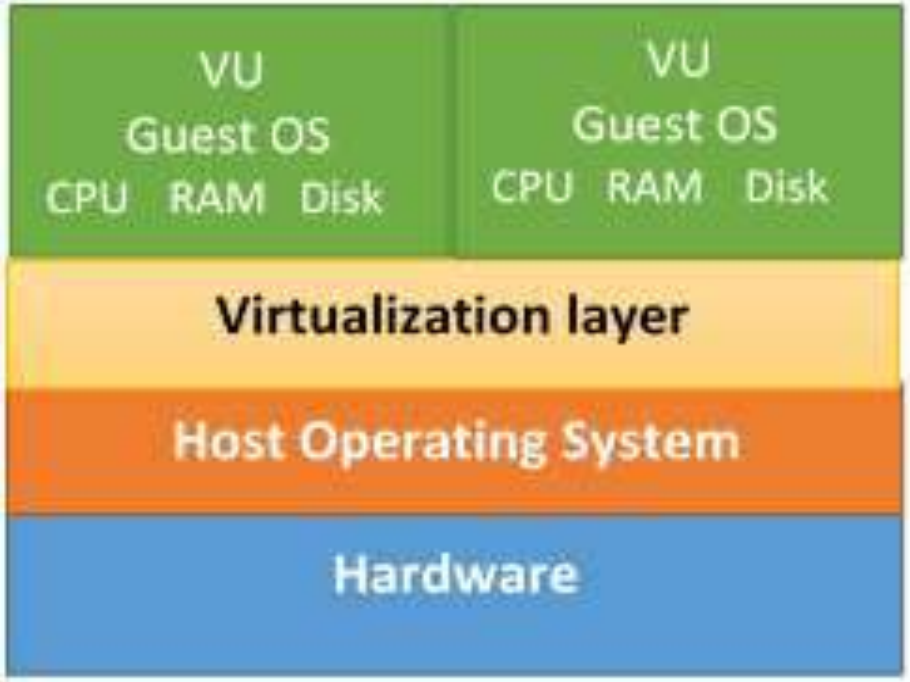
- What is Virtual Machine?
- Uses of Virtual Machine
- Virtual Machine's Files

Virtual Machine



- Virtual Machine is a virtual representation of a Physical machine
- It is called as Guest machine or Virtual Server
- It provides same functionalities as Physical machine or Server
- It uses the resources of physical machine through virtualized software
- Each Virtual Machine has its own operating system called as Guest Operating System

Virtual Machine



Uses of Virtual Machine



- VM uses as a substitute of Physical Server in a Data Center

replacement

- Reduce Data Center Cost

Hardware Reduction: Multiple VMs run on a single physical server, reducing the need for more hardware.
Lower Power and Cooling Costs: Fewer physical servers mean less electricity and cooling required.
Reduced Maintenance Costs: Maintaining VMs is cheaper than maintaining physical hardware.

- VM provides different types of Virtualization like Hardware virtualization, desktop virtualization etc.
- Provides Scalability (add new VM or upgrade existing one)
- Migration is quite easier

Virtual Machine Files

File	Brief Description
vmname.nvram	VM BIOS Configuration file
.vmx	Configuration file of VM. It contains information of hardware setting
.vmsn	VM snapshot data files
.vmem	Virtual Machine Paging backup file
.vmdk	Virtual disk characteristics
vmware.log	Stores the log activities of VM

Virtual Networking



What is Virtual Networking

- **Topics to be Covered**:
  - Creation of virtual versions of networking components
    - Like virtual switches, virtual routers, and virtual firewalls
  - Enables the use of software-based solutions for networking tasks
  - Simplifies network management and improve network agility

What is Virtual Networking...

Virtual Networking ka matlab hai software ke through network create karna aur manage karna, bina physical devices (switches, routers) ke.

- Allows data centers to stretch across different physical locations
- Enables Network Administrators to easily modify the network as required
- No need to switch out or buy more hardware
- Greater flexibility in provisioning the network to specific needs

Ek company ka ek office Mumbai mein aur doosra Delhi mein hai. Virtual networking ke zariye, dono offices ke servers ko aise connect kiya ja sakta hai ki employees ko files aur applications access karne mein koi dikkat na ho.

: Network mein changes karna, jaise naye devices add karna ya nayi settings configure karna, bohot easy hota hai. Physical wiring ya hardware ki zarurat nahi hoti.

Virtual networking se naye hardware kharidne ki zarurat nahi hoti, aur network ko fast aur cost-effective tareeke se upgrade kar sakte hain.

Virtual networking specific zarurat ke hisaab se network ko customize karne ki flexibility deta hai, jaise data ko isolate karna ya kisi application ke liye zyada bandwidth allocate karna

Tools of Virtual Network



Summary

- **Router**: Different networks ko connect karne aur internet access provide karne ke liye.
- **Switch**: Local devices ko ek hi network mein efficiently connect karne ke liye.

Tools of Virtual Network:

Virtual switches enable communication between VMs and physical networks.
Virtual routers connect virtual subnets and enforce routing/security policies.
Virtual firewalls secure virtual networks by controlling and protecting data traffic.

Copy

- **Virtual switches:** Emulate physical switches and allow virtual machines to communicate with each other and with the physical network.
- **Virtual routers:** Provide routing capabilities between virtual subnets and can also be used to implement network security policies.
- **Virtual firewalls:** Offer firewall protection for virtual networks and can be used to enforce security policies.

Tools of Virtual Network:

- **Virtual load balancers:** Distribute network traffic across multiple virtual machines to improve performance and availability.
- **Virtual VPNs:** Provide secure remote access to virtual networks from remote locations.
- **Virtual network adapters:** Connect virtual machines to virtual networks, just like physical network adapters connect physical machines to physical networks.

Tools of Virtual Network:

- **Virtual network security groups:** Provide security for virtual networks by defining access controls for incoming and outgoing network traffic.

- **Virtual network monitoring tools:** Monitor virtual network performance, identify network issues and provide network visibility.

Virtual Load Balancers: Distribute traffic to improve performance and availability.
Virtual VPNs: Ensure secure remote access to virtual networks.
Virtual Network Adapters: Enable VMs to connect to virtual networks, similar to physical adapters.

Virtual Network Security Groups: Protect virtual networks by managing traffic with access control rules.
Virtual Network Monitoring Tools: Ensure optimal performance and network visibility by tracking and troubleshooting issues in real time.

Virtual Switch & its Types



Virtual Switch

- Same as physical switches connect physical machines
- Software-based switch that provides network connectivity to VMs
- Allows virtual machines to communicate with each other
- Can be created and managed within a virtualization platform

Types of vSwitch

- There are there types vSwitches provided by Hyper-V

- External vSwitch
- Internal vSwitch
- Private vSwitch

Feature	External vSwitch	Internal vSwitch	Private vSwitch
Network Connectivity	Connects VMs to the **physical network** (internet, external resources)	Connects VMs to the **Hyper-V host** and each other	Connects only **VMs to each other** (no access to physical network or host)
Access to Physical Network	Yes, allows access to external resources (internet, LAN)	No, VMs cannot access external resources	No, VMs are isolated from external networks entirely
Communication Between VMs	Yes, VMs can communicate with each other through the external network	Yes, VMs can communicate with each other **and** with the host	Yes, but only VMs on the same Hyper-V host, and **not** the host itself

Internal vSwitch

- Provides communication between virtual machines and the Hyper-V host
- Used to isolate virtual machines from each other
- Allow communication within the same host

Private vSwitch

- Provides network connectivity only between virtual machines running on the same hypervisor host.
- Does not provide access to the physical network
- Used to isolate virtual machines from each other for security or testing purposes

External vSwitch

- Provides network connectivity between virtual machines and the physical network.
- Used for virtual machines that need access to the physical network

Standard Switch Architecture



Standard Switch Architecture...

- Refers to the traditional design and structure of a network switch. The architecture typically consists of the following components:

- Input/Output (I/O) interfaces
- Switch Fabric
- Processor
- Memory
- Power supply
- Operating system
- Management interface

Standard Switch Architecture...

- **Input/Output (I/O) interfaces**: The ports on a switch where data enters and exits the switch.
- **Switch Fabric**: The internal component of a switch responsible for forwarding and switching packets between the I/O interfaces.
- **Processor**: The component of a switch responsible for managing the switch and its configuration, as well as performing packet forwarding decisions.

Standard Switch Architecture...

- **Memory**: Used to store the switch's configuration and forwarding information.
- **Power supply**: Provides power to the switch components.
- **Operating system**: The software that runs on the switch

Standard Switch Architecture...

- **Management interface:** The interface used to configure and manage the switch, such as a command-line interface (CLI) or web-based graphical user interface (GUI).

Central Aspects of Network Security and Compliance



Central Aspects of Network Security and Compliance



1. Network Policies
2. Security
3. Traffic shaping
4. NIC teaming
5. Failover

Central Aspects of Network Security and Compliance

- **Network policies:** A set of rules and procedures that govern the use of the network and help ensure that data and devices are protected.
- **Security:** Measures such as firewalls, intrusion detection and prevention systems, encryption, and access controls to prevent unauthorized access and protect against security threats.
- **Traffic shaping:** Techniques used to manage and control the flow of network traffic to ensure that critical applications receive sufficient bandwidth and to prevent network congestion.

Central Aspects of Network Security and Compliance

- **NIC Teaming:** The process of combining multiple network interfaces into a single logical interface to provide increased bandwidth, failover protection, and load balancing.

- **Failover:** A backup solution that provides automatic failover to an alternate system or device in the event of a failure or outage.



Overview of Virtual Storage

Overview of Virtual Storage

- Refers to the creation of a virtual representation of storage devices
- a software layer acts as an intermediary between the operating system and the physical storage
- enables the creation of multiple virtual storage devices from a single physical storage resource
- Provides improved resource utilization, enhanced data protection, and simplified data management.

Overview of Virtual Storage

Pizza Dough example

- **Improved resource utilization:** Virtual storage enables the efficient use of storage resources by pooling physical storage and allocating it to virtual storage devices as needed.Akhataa kerna / Jamaa kernaa

- **Enhanced data protection:** Virtual storage provides features such as data snapshots, replication, and backups that help protect data and ensure its availability.

Overview of Virtual Storage

- Simplified data management: Virtual storage abstracts the physical storage, making it easier to manage, provision, and allocate storage resources. This reduces the time and effort required for tasks such as adding or removing storage, or resizing storage devices.

Ability to grow

- Increased scalability: Virtual storage makes it easier to scale storage resources as needed, allowing for the seamless addition of storage capacity as required.

Overview of Virtual Storage

- **Flexibility:** Virtual storage enables the creation of multiple virtual storage devices, each with its own characteristics, such as size, performance, and availability, from a single physical storage resource.

- **Improved performance:** Virtual storage can improve the performance of storage-intensive applications by providing features such as caching, data compression, and de-duplication.

Overview of Virtual Storage

- Improved disaster recovery: Virtual storage makes it easier to create and manage disaster recovery solutions, as virtual storage devices can be quickly moved to alternate locations or stored in the cloud for easy access in the event of a disaster.
- **Simplified data management:** Virtual storage abstracts the physical storage, making it easier to manage, provision, and allocate storage resources. This reduces the time and effort required for tasks such as adding or removing storage, or resizing storage devices.



Overview of Virtual Storage

Overview of Virtual Storage



- **Relational databases:** These store data in tables with rows and columns and use structured query language (SQL) to access and manipulate data. Examples include MySQL, Oracle, and Microsoft SQL Server.

- **NoSQL databases:** These databases store data in a non-relational format, such as key-value pairs, document-based, or graph-based. NoSQL databases are often used for handling large amounts of unstructured or semi-structured data. Examples include MongoDB, Cassandra, and CouchDB.

Overview of Virtual Storage

- **Object-oriented databases:** These store data as objects, which are instances of classes, rather than as tables with rows and columns. Examples include MongoDB and CouchDB.

- **Time-series databases:** These databases are optimized for storing and querying time-stamped data. Examples include InfluxDB and OpenTSDB.

Overview of Virtual Storage

- **Columnar databases**: These databases store data in columns rather than rows, allowing for faster data retrieval for certain types of queries. Examples include Apache Cassandra and Hbase.

- **Key-value stores**: These databases store data as a collection of key-value pairs, where each key is unique and is associated with a value. Examples include Redis and Riak.

Overview of Virtual Storage

- **File-based databases:** These databases store data as files on a file system. Examples include JSON or XML files.

Full Virtualization vs Para Virtualization



Full Virtualization vs Para Virtualization

Full Virtualization

- In Full Virtualization, the Guest OS in a VM is completely decoupled from the underlying hardware and Virtualization layer
- In this technique, the unmodified Guest OS is installed.

Para Virtualization

- In Para Virtualization, the Guest OS in a VM is partially isolated from the underlying hardware and Virtualization layer
- In this technique, the modified Guest OS is installed.

Full Virtualization vs Para Virtualization

Full Virtualization

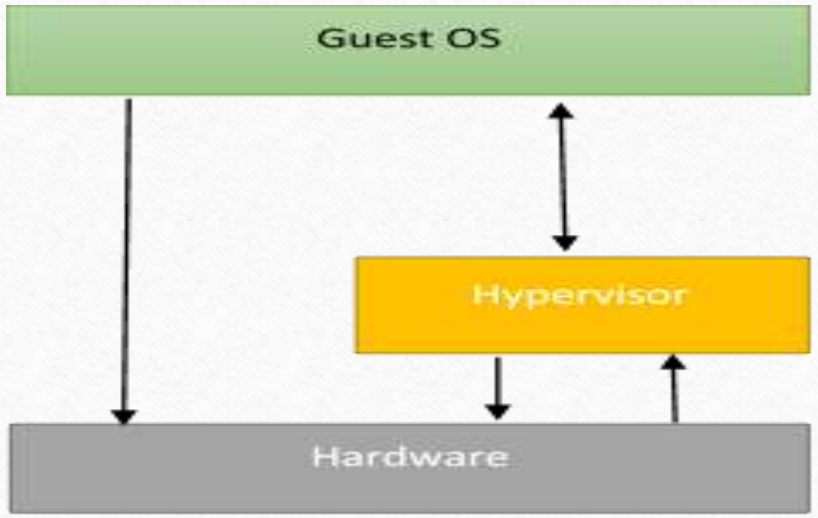
- It allows the Guest OS to execute independently.
- The Guest OS is unaware of the virtualization
- In this technique, Guest OS issues hardware calls to access hardware

Para Virtualization

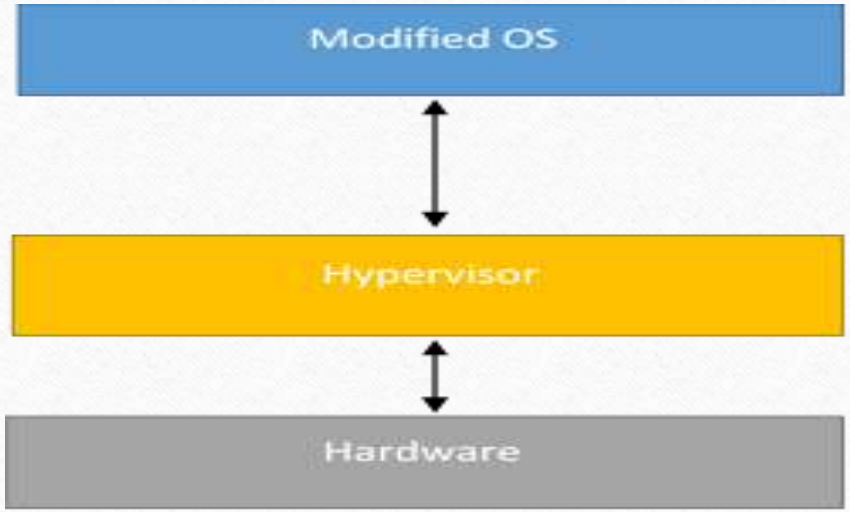
- It allows the guest OS to interact with the hypervisor
- The Guest OS is aware that it is being virtualized
- In this technique, Guest OS issues hyper calls to hypervisor to access hardware

Full Virtualization vs Para Virtualization

Full Virtualization



Para Virtualization



Full Virtualization vs Para Virtualization



Full Virtualization

- In Full Virtualization, performance is low.
- Full virtualization is less secure
- It is more portable and compatible

Para Virtualization

- In Para Virtualization, performance is comparatively high.
- Para virtualization is more secure
- It is less portable and compatible

Full Virtualization



Full Virtualization



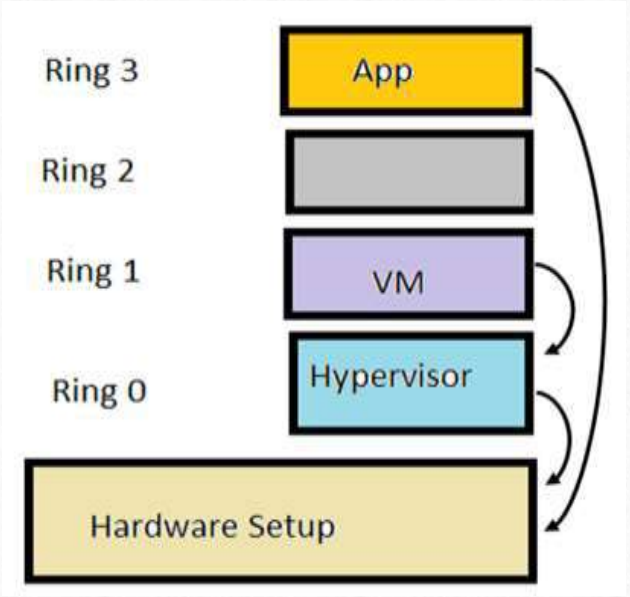
- Full Virtualization is a virtualization technique in which VMs or Virtual Servers are created and run on a single physical server.
- Each VM has its own OS and runs independently
- The software layer providing the virtualization is called as Hypervisor or Virtual Machine Monitor (VMM).
- The hypervisor is inserted b/w the hardware and VMs

Full Virtualization

Virtual University

Separated

- The guest OS is completely decoupled from the underlying hardware
- Each guest OS is unmodified and unaware that it is being virtualized
- The application/user level instructions are directly executed by underlying hardware
- The OS level requests are translated by Hypervisor at run time



Full Virtualization

- Examples of full virtualization are VMware ESX$_i$, Virtual Box and Microsoft Virtual Server.

Disadvantages



- The performance of full virtualization may not be ideal as the Binary translation of OS requests takes place at run time
- Any physical server fault or failure can effect all the VMs

Para Virtualization



Virtaul System and Services

Para Virtualization



- Para Virtualization is a virtualization technique, used to address the issues found in full virtualization
- Like Full virtualization, this technique allows us to create and run multiple VMs on a single physical server.
- In this technique, the Guest OS in not completely isolated from the under lying hardware and Virtualization layer

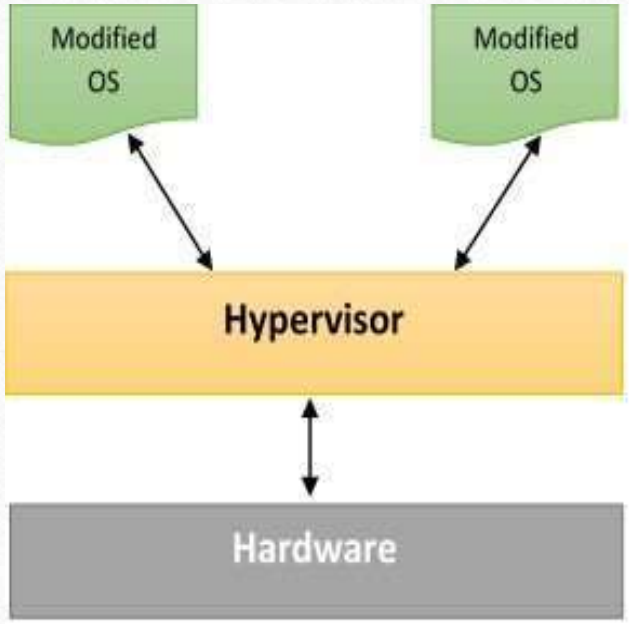
Para Virtualization



- The Guest OS is modified to work with a VM. A special software called drivers are installed with each guest OS.
- The guest OS is aware that it is running in virtual environment on top of hypervisor
- The hardware resources are shared among the Guest OSs instead of emulating the whole hardware

Para Virtualization

- Instead of issuing calls to native hardware, the Guest OS issues hyper calls to hypervisor
- The Hypervisor handles hyper calls at compile time and then perform operation on hardware.
- The result is communicated back to VM through hypervisor



Para Virtualization



- **Advantages**:
  - Virtualization overhead reduced
  - Performance improved
- **Disadvantages**:
  - Modified Guest OS is required to work with hypervisor
  - Limiting support to open source OSs like Linux
  - Less portable and compatible

Hints

- Para Virtualization cannot work with unmodified OS.
- Xen an open source hypervisor
- **Virtualization overhead reduced**: hypervisor has less responsibility; instead of emulating the whole hardware it receives hyper call & perform operation on hardware

Containers



Virtual Systems and Services

Container



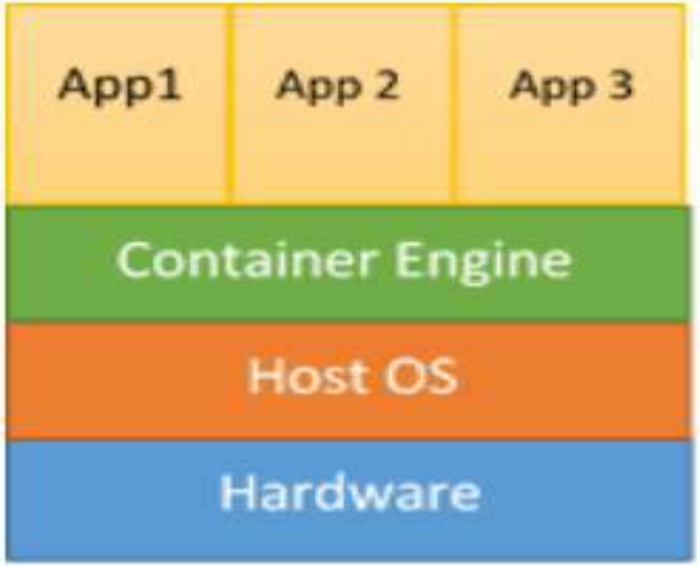
- A Container is a software unit in which Application code, its libraries and dependencies are packaged together
- Unlike Virtual machine, it doesn't include Operating System
- It is a small, fast and light weight unit
- It sits on top of physical server and virtualizes the OS only.

Container



- It accesses the operating system kernel via container engine
- It uses the physical resources of hosting machine through container engine
- It is portable unit because each container image includes all the dependencies to execute
- Multiple independent containers can run on a single OS
- Cloud applications are built on containers

Container



Container Software



- Docker
- Microsoft Azure
- Amazon ECS
- Google Cloud Platform

Merits and Demerits of Container



Merits of Container

- **Containers are most light weight**
  - Their image include only the application related data (in MB instead of GB)
- **Less overhead**

Ksi chz ko active kerna

  - Requires less system resources to deploy, run and manage
- **Boot Fast:** takes less time in boot up due to less overhead

On kernaa

- **Operational Efficiency**
  - Operationally efficient than VM

Merits of Container



- **Portability**
  - Container encapsulates all dependencies, own runtime environment; runs on different platforms. It is write once and runs anywhere
- **Isolation**
  - Containers run in isolation
- **Security**
  - Security improve due to isolation, each container runs in self-contained environment

Demerits of Container



- **Need Dependencies**
  - Packaging workload with supporting dependencies is one of the limitations.
- **Issue with Stateful Applications**
  - It is difficult to containerize stateful applications. They need parallel system to run like database access.

Jo apna data store / save rkhti ha

- **Security**
  - Since application layer and host OS is shared across the containers, it opens the door to security breaches across containers

Virtualization vs Containerization



Virtualization vs Containerization



- Virtualization makes use of Virtual Machines to convert a traditional hardware into a virtual environment
- Virtualization allows us to run multiple virtual servers on top of a single physical machine

- Containerization makes use of containers to convert a traditional hardware into a virtual environment
- Containerization allows us to run multiple applications on top of a single physical machine

Virtualization vs Containerization



- Virtualization enables us to efficiently utilize the computer's hardware resources
- Virtualization uses Virtualized software to create an abstract layer over hardware

- Containerization enables us to efficiently utilize the computer's host operating system
- Containerization uses Container Engine to create an abstract layer over hardware

Virtualization vs Containerization



- Virtualization allows developers to create and deploy VMs faster & more securely
- In Virtualization, an OS is a part of VM
- Use of Virtual Servers or VMs reduces the hardware cost

- Containerization allows developers to create and deploy applications faster & more securely
- In containerization, an OS is not a part of container
- Use of Containers reduces the software cost

Virtualization vs Containerization



- VMs take more time to boot up
- Containers take less time to boot up